

MOSAIC: Multi-Objective Search for Allocating Inference Cache Across LLM Decoder Layers

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Abstract

The Key-Value (KV) cache is the main memory bottleneck of long-context LLM inference. Token eviction methods such as SnapKV, H2O, AdaKV and L2Norm shrink it, but they give every transformer layer the same budget, although layers differ in which tokens they need. We present MOSAIC, a multi-objective neural architecture search that learns per-layer KV-cache budgets for any eviction method. One search maps the trade-off between average budget and task score; its best allocation transfers to every target budget without further search, and an optional budget-constrained search refines it. Across four eviction methods on Llama-3-8B-Instruct and Mistral-7B-Instruct-v0.2, MOSAIC beats uniform allocation in **146 of 154** (task, method, budget) settings, by up to **+48 points** on RULER and **+11.9** on LongBench (LCC). It reaches the score of uniform allocation with up to **4× less KV cache (75% smaller)**, and with half the cache or less in 17 Llama settings. The transferred allocation alone beats uniform in **75 of 87** Llama settings, so **a single 9–40 GPU-hour search serves every budget**. Against EvolKV, MOSAIC explores a $\sim 10^{40} \times$ larger search space with **4× fewer evaluations** and **18% fewer GPU-hours** per budget, and scores higher in **5 of 6** settings. The learned allocations consistently favour a small set of middle layers.

1 Introduction

Long-context inference with large language models (LLMs) is limited by the memory of the key-value (KV) cache, which grows linearly with sequence length and batch size (Kwon et al., 2023): a 30B-parameter model with batch size 128 and sequence length 1024 already needs 180 GB of KV cache (Zhang et al., 2023). Token eviction methods like H2O (Zhang et al., 2023), SnapKV (Li et al., 2024) and L2Norm (Devoto et al., 2024) address this by selectively dropping cache entries based on token

importance, and AdaKV (Feng et al., 2025) builds on them by distributing a layer’s budget adaptively across its attention heads. However, these methods apply the same memory budget to every transformer layer. Layer-wise allocation schemes exist, but they either fix the shape of the allocation in advance (Cai et al., 2025), derive it from attention statistics (Qin et al., 2025), or optimize layer budgets one group at a time at one budget and rescale the result to others (Yu and Chai, 2025). These schemes already exploit layer differences; MOSAIC instead searches all layers jointly once and transfers the result to every budget. In our searches, a few middle layers consistently receive more than their uniform share, so a uniform allocation spends cache on layers that need little while starving the ones that need more.

This paper introduces MOSAIC, a method that automatically optimizes per-layer KV-cache budgets under a target memory constraint using multi-objective neural architecture search. Rather than hand-tuning layer-specific heuristics or optimizing layers group by group, MOSAIC searches the per-layer budget space jointly, one budget per decoder layer (32 for Llama-3-8B, up to 10^{27} configurations on its search grid), using Bayesian optimization guided by a surrogate MLP (Figure 1). MOSAIC is to KV-cache allocation what static quantization is to model quantization: just as post-training quantization calibrates offline on a small calibration set (Frantar et al., 2023; Xiao et al., 2023), MOSAIC fixes each layer’s budget offline on calibration data, so inference adds no per-prompt allocation step. Methods such as CAKE and Ada-KV instead adapt the budget to each prompt at inference time, as dynamic quantization computes its ranges at run time. We evaluate MOSAIC across four eviction methods (SnapKV, H2O, AdaKV, L2Norm) on Llama-3-8B-Instruct and Mistral-7B-Instruct-v0.2 (all four methods; on LongBench Summarization, SnapKV, H2O and AdaKV ran Stage 1

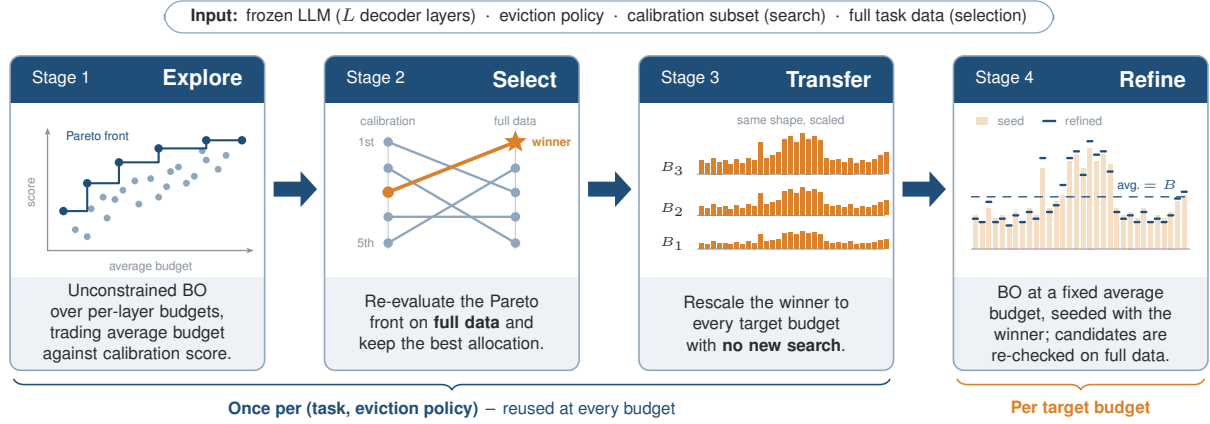


Figure 1: **The MOSAIC pipeline.** Stages 1–3 run once per (task, eviction policy): an unconstrained search on a calibration subset yields a Pareto front of per-layer allocations, full-data re-evaluation selects a winner, and the winner is rescaled to every target budget without further search. Stage 4 optionally refines each budget with constrained BO seeded by the rescaled winner. Bar profiles are schematic.

only), on two benchmarks: LongBench (4 task categories) and RULER (11 in-context retrieval subtasks). On RULER, per-layer allocation yields large gains over uniform (+2.3 to +47.9 points), concentrated on the hardest needle-retrieval subtasks. On LongBench, MOSAIC improves every Code setting, with dataset-level gains of up to +11.9 points on LCC (L2Norm), +3.5 on LCC for SnapKV, and +3.0 on MultiFieldQA-en (H2O) and Qasper (AdaKV) (Section 4.2).

MOSAIC versus EvolKV. The closest prior work, EvolKV (Yu and Chai, 2025), searches per-layer budgets with CMA-ES (Hansen and Ostermeier, 2001) group by group, eight layers at a time, at a target budget fixed in advance. MOSAIC instead searches all layer budgets jointly, so one search spans about 10^{40} times more allocations than one EvolKV group search, across a budget–score Pareto front, selects a reusable anchor, and optionally refines it at each deployment budget; on Code it beats EvolKV at every budget, both searching once (+1.3 to +2.1) and per budget at EvolKV’s compute (+0.8 to +1.7, Appendix D), and even searched once, MOSAIC’s winner outperforms EvolKV’s per-budget searches at every budget where both ran (Section 6).

Our contributions are: (i) MOSAIC, a four-stage workflow (joint per-layer search, budget–score exploration, winner rescaling, optional per-budget refinement) whose single search yields a winner that transfers across budgets without further search; (ii) an evaluation across four eviction methods, two models and two benchmarks, where MOSAIC improves over uniform allocation for every method,

with gains of up to +48 points on RULER; and (iii) an analysis of the learned allocations, which favour a small set of middle layers.

2 Related Work

Token eviction. Eviction methods reduce KV-cache memory by retaining only the tokens judged important. H2O (Zhang et al., 2023) keeps “heavy hitters”, tokens with high accumulated attention, together with a window of recent tokens, as do StreamingLLM (Xiao et al., 2024), Scissorhands (Liu et al., 2023) and FastGen (Ge et al., 2024) with sink, persistent and per-head criteria. SnapKV (Li et al., 2024) selects, for each attention head, the prompt positions that receive the most attention from an observation window at the end of the prompt. L2Norm (Devoto et al., 2024) keeps the tokens whose keys have the lowest L_2 norm, which correlates with high attention, without computing attention scores. KeyDiff (Park et al., 2025) also avoids attention scores, keeping keys that are geometrically distinctive from the others, which makes it compatible with FlashAttention (Dao, 2024). QUOKA (Jones et al., 2026) selects key-value pairs for prefill from the keys aligned with a subset of representative queries. These methods decide which tokens to keep; how much cache each layer should receive is a separate question, which MOSAIC addresses.

Static and dynamic budget allocation. A second line of work allocates a fixed total budget non-uniformly instead of changing how tokens are scored. Dynamic methods recompute the allocation for every prompt, as dynamic quantization com-

puts its ranges at run time: Ada-KV (Feng et al., 2025) allocates budget adaptively across the heads of a layer, CAKE (Qin et al., 2025) derives per-layer budgets from layer preferences measured on each prompt’s attention patterns, and SqueezeAttention (Wang et al., 2025) shifts budget between two layer groups. Static methods fix the allocation before inference: PyramidKV (Cai et al., 2025) assigns layer budgets following a hand-designed pyramidal shape, larger in lower layers and smaller in higher ones; PyramidInfer (Yang et al., 2024) keeps per layer the context attention consistently selects, less with depth; HeadKV (Fu et al., 2025) and DuoAttention (Xiao et al., 2025) fix head-level allocations offline. MOSAIC is static in the sense of static quantization, calibrating the allocation offline for each model and eviction method. Closest to our work, EvolKV (Yu and Chai, 2025) treats per-layer budgets as optimization variables, partitions layers into groups, and runs evolutionary search on one group at a time against downstream task score at one target budget; its main results then extend that allocation to larger budgets by proportional rescaling, without further optimization. MOSAIC shares this rescaling step but finds the allocation to rescale differently, searching all layers jointly across budgets rather than at one target budget (Section 6).

Neural architecture search. Multi-objective NAS trades off competing goals such as accuracy and latency along a Pareto front (Elsken et al., 2019) (more related work in Appendix G). MOSAIC adopts this multi-objective formulation for KV-cache allocation but uses its own search procedure (Section 3): a classifier-guided optimizer built for this problem, which needs only a few hundred evaluations per search.

3 Method

3.1 Problem Formulation

Given an eviction method M , a model with L decoder layers ($L = 32$ for Llama-3-8B) and a target budget B (average tokens per layer), we seek a vector $\mathbf{b} = (b_1, \dots, b_L) \in [b_{\min}, b_{\max}]^L$ that maximizes task performance while maintaining $\frac{1}{L} \sum_{i=1}^L b_i = B$. We use a per-layer floor $b_{\min} = 64$ (Appendix E) and cap $b_{\max} = 4096$. Each configuration is encoded as $\mathbf{x} \in [0, 1]^L$. The unconstrained search is multi-objective: it minimizes the mean per-layer budget $f_1(\mathbf{b}) = \frac{1}{L} \sum_i b_i$ and maximizes the task score $f_2(\mathbf{b})$ on a calibration

subsample. The budget-constrained variant (Stage 4) instead fixes $f_1 = B$ by rescaling the per-layer weights so that their mean equals B exactly.

Search space. Stage 1’s grid of $|G|$ levels per layer holds $|G|^L$ configurations ($5^{32} \approx 2.3 \times 10^{22}$ or $7^{32} \approx 1.1 \times 10^{27}$ for $L = 32$) spanning every average budget from 64 to the grid cap, and at a fixed budget B the integer allocations in $[64, 4096]^L$ with mean B number about 10^{69} at B128 and 10^{105} at B1024. One EvolKV search varies only eight layers with the other $L - 8$ fixed: even ignoring its budget penalty, that is at most $4033^8 \approx 10^{29}$ allocations, about 10^{40} times fewer than one MOSAIC search at B128 (Appendix H). A search evaluates at most 200 of them in Stage 1 and 150 in Stage 4 (Appendix H), so every layer budget must be chosen jointly from a vanishing fraction of the space.

Search step. All stages share one classifier-guided Bayesian optimization procedure (cf. Tiao et al., 2021). After each evaluation, the archive is non-dominated sorted (Deb et al., 2002), an MLP classifier (two hidden layers of 32 units) is trained to separate the rank-1 configurations from the rest, and differential evolution (Storn and Price, 1997) maximizes the classifier’s probability of rank 1 over $[0, 1]^L$ to propose the next configuration, one evaluation per iteration. We call these BO proposals; stopping rules are in Appendix H.

3.2 Search Procedure (MOSAIC — Four-Stage Pipeline)

Why four stages. An unconstrained search returns configurations at arbitrary average budgets, so a configuration it finds cannot be compared directly with uniform allocation at a fixed budget B . MOSAIC therefore selects one winner shape from the search (Stages 1–2) and rescales it to each target budget (Stage 3). This already gives a configuration at every budget without further search, and the pipeline can stop here. Stage 4 is optional: it searches at a fixed budget for a configuration better than the rescaled winner (Figure 1).

Tasks. MOSAIC searches separately for each task and eviction method. LongBench has four English task categories, each a group of datasets: Single-Doc QA (NarrativeQA, Qasper, MultiFieldQA-en), Multi-Doc QA (HotpotQA, 2WikiMultihopQA, MuSiQue), Summarization (GovReport, QMSum, MultiNews) and Code (LCC, RepoBench-P); a category’s score is the mean over its datasets. We search each LongBench category separately because their tasks and metrics

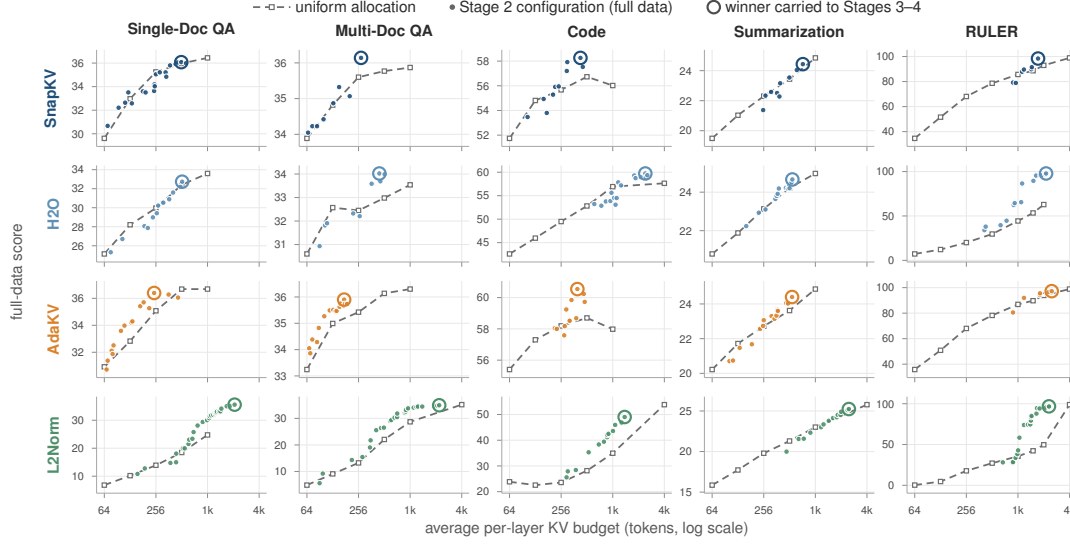


Figure 2: **Budget–score trade-off found by the unconstrained search (full data).** Rows: eviction method; columns: task. Dashed: uniform allocation. Dots: configurations on the Stage 1 Pareto front, re-evaluated on full data in Stage 2. Ring: the winner carried to Stages 3 and 4.

differ (F1 for QA, ROUGE-L for summarization, edit similarity for code). RULER’s 11 retrieval subtasks at 4K context share one metric (string-match accuracy), so a single search covers all of them. Stages 1–2 run once per (task, method) and Stages 3–4 once per target budget; we call each (task, method, budget) a *cell*.

Stage 1 (Unconstrained Multi-Budget NAS):

Search a discrete per-layer grid of powers of two with no budget constraint, from 64 up to 4096 on RULER and up to 1024 or 4096 on LongBench (Appendix H). The initial design has 64 points: the uniform allocation at every grid level plus Latin Hypercube samples (McKay et al., 1979). Guided iterations then trace out the budget–score Pareto front. Because this search must rank allocations across the whole budget range, and runs only once per (task, method), it uses a larger calibration sub-sample on LongBench (30%; case study in Appendix B); on RULER it uses 10%, which already ranks faithfully (Appendix B).

Stage 2 (Winner Selection via Full-Data Eval):

Re-evaluate every configuration on the Stage 1 Pareto front (typically 10–20 configurations) on the full test data, and keep the **winner shape**: the shaped configuration with the highest full-data score. This winner becomes a universal anchor for all subsequent budget-constrained searches.

Stage 3 (Winner Rescaling to Fixed Budgets):

Take the winner shape discovered in Stage 2 and rescale it to match fixed target budgets: B128–

B1024 on LongBench and B64–B2048 on RULER (B64 only where the floor of 16 allows reallocation). Evaluate each rescaled configuration on full test data. These Stage 3 results (**Winner-Rescaled**) require no further search, and they already beat uniform in 75 of 87 (category, method, budget) cells, including shapes found at an average budget of ~ 2400 transferred down to 128. Table 1 summarizes the transfer; all 87 cells are in Appendix A.

Stage 4 (Budget-Constrained Adaptive Search):

For each target budget B , Stage 4 solves $\max_{\mathbf{x} \in [0,1]^L} f_2(D_B(\mathbf{x}))$, where the decoder D_B maps $\mathbf{x}$ to budgets with $\frac{1}{L} \sum_i b_i = B$ and $b_{\min} \leq b_i \leq b_{\max}$: it sets $b_i \propto x_i$ scaled to the total LB , clamps layers that cross a bound and redistributes the remainder in proportion to x_i , then rounds so that the total is exactly LB (Appendix H). Every candidate thus meets the budget by construction, and with f_1 fixed at B the search maximizes f_2 alone. Seed this search with: (i) the Stage 2 winner rescaled to budget B (from Stage 3), (ii) six heuristic shapes (uniform, ascending and descending ramps, middle- and edge-heavy triangles, and an alternating low/high pattern), and (iii) 57 Latin Hypercube (LHS) seed points, all drawn inside the constrained per-layer space (continuous weights rescaled to mean B). Guided BO iterations then extend this pool; run lengths varied across methods and cells, up to a cap of 180/150 evaluations (stopping rules in Appendix H). Stage 4 runs once for every target

Task	Method	Stage 3 (Winner)		Stage 4 (NAS-refined)		MOSAIC (best of both)	
		Wins	Δ	Wins	Δ	Wins	Δ
Code	H2O	3/4	+0.54	4/4	+1.00	4/4	+1.07
Code	SnapKV	4/4	+1.66	4/4	+1.77	4/4	+1.81
MultiDoc	H2O	3/4	+0.55	3/4	+0.46	3/4	+0.70
MultiDoc	SnapKV	3/4	+0.31	3/4	+0.06	3/4	+0.32
SingleDoc	H2O	3/4	+0.66	4/4	+0.64	4/4	+0.99
SingleDoc	SnapKV	3/4	+0.04	2/4	+0.04	3/4	+0.22
Summ	H2O	4/4	+0.11	4/4	+0.27	4/4	+0.28
Summ	SnapKV	4/4	+0.19	0/1	-0.08	1/1	+0.12
RULER	H2O	6/7	+7.86	4/4	+21.84	4/4	+21.84
RULER	SnapKV	7/7	+6.57	5/5	+5.83	5/5	+6.97
Code	AdaKV	2/3	+0.38	3/3	+1.47	3/3	+1.47
MultiDoc	AdaKV	2/3	+0.22	3/3	+0.59	3/3	+0.62
SingleDoc	AdaKV	2/3	+0.12	2/3	+0.84	2/3	+0.84
Summ	AdaKV	1/3	-0.02	3/3	+0.36	3/3	+0.36
RULER	AdaKV	7/7	+2.16	6/6	+5.90	6/6	+5.90
Code	L2Norm	4/4	+6.87	4/4	+6.87	4/4	+7.33
MultiDoc	L2Norm	3/4	+2.19	4/4	+3.29	4/4	+3.29
SingleDoc	L2Norm	4/4	+1.90	4/4	+1.75	4/4	+2.10
Summ	L2Norm	4/4	+0.38	4/4	+0.44	4/4	+0.59
RULER	L2Norm	6/6	+5.49	6/6	+23.54	6/6	+23.54
Total		75/87		72/78		74/78	

Table 1: **One search, many budgets.** Stage 3: the Stage 2 winner, found once at its natural average budget and rescaled to every target budget with no further search. Stage 4: NAS-refined, the best configuration of the budget-constrained search, available only at budgets where it ran. MOSAIC: its final configuration, the better of Winner and NAS-refined (chosen per budget), at the same budgets as Stage 4. Wins: budgets where the configuration beats uniform on full data; Δ : mean gain over uniform. Llama-3-8B-Instruct; Mistral-7B-Instruct-v0.2 is in Table 9.

budget and evaluates many configurations each time, so it uses a smaller subsample (10% on both benchmarks) to keep each per-budget search affordable. Its best candidates are re-evaluated on full data, and the best of them is reported as **NAS-refined** (candidate selection in Appendix H; calibration fidelity in Appendix B). For each budget we thus report uniform, the rescaled Stage 2 winner (**Winner**), and NAS-refined; **MOSAIC’s final configuration is the better of Winner and NAS-refined on full data** (Section 4.2). Stage 4 does not always improve on the Winner: of the 78 cells, NAS-refined matches or beats the Winner in 54, and the Winner remains better in 24 (Figure 4).

Selection and evaluation data. The *full data* of a cell are all examples of its task’s datasets (150–500 per LongBench dataset, 500 per RULER subtask); the *calibration subset* is a fixed random fraction of each dataset (30% or 10%, as above). Search uses the calibration subset; selection in Stages 2 and 4 and all reported scores use the full data. LongBench and RULER have no validation split, so configurations are selected and scored on the same examples: our results show gains on the

evaluated examples, not generalization to unseen ones.

4 Experiments

4.1 Setup

Models: Meta-Llama-3-8B-Instruct (Grattafiori et al., 2024) (all four methods) and Mistral-7B-Instruct-v0.2 (Jiang et al., 2023) (all four methods; on LongBench Summarization, SnapKV, H2O and AdaKV ran Stage 1 only). **Benchmarks:** LongBench (Bai et al., 2024) and RULER (Hsieh et al., 2024) (tasks and data in Section 3.2). **Methods:** SnapKV, H2O, AdaKV, L2Norm. **Budgets:** average per-layer budgets of 128–1024 on LongBench and 64–2048 on RULER; each layer receives between 64 and 4096 tokens (16 in some earlier RULER runs; Section 4.3). **Search signal:** search-time scores use a calibration subsample drawn with a fixed seed: on LongBench 30% of each dataset in Stage 1 and 10% in Stage 4, and on RULER 10% of each subtask in both stages, for all four methods on Llama; all Mistral LongBench and L2Norm searches use 30% in every stage (Mistral H2O truncates prompts at 7.5K tokens, Appendix A). Hardware is listed in Appendix H.

4.2 Key Findings — LongBench

Figure 2 shows, for every eviction method and task, the budget–score trade-off that the unconstrained search finds. The dots are the configurations on the Stage 1 Pareto front, which spans a wide range of average budgets within a single search, re-evaluated on full data in Stage 2; the ring marks the Stage 2 winner. Among the strongest Stage 1/2 results, SnapKV’s Code winner (58.26 at average budget 430) beats uniform at every budget up to 1024 (at most 56.73), and its Multi-Doc QA winner (36.14 at 274) beats uniform-1024 (35.87) with 27% of its cache.

Where MOSAIC loses. Table 2 gives the full LongBench results. Across all evaluated cells, MOSAIC beats uniform in 74 of 78 (SnapKV and H2O: 35/38; AdaKV: 17/18; L2Norm: 22/22), including every CODE, Summarization and RULER cell. On Mistral-7B-Instruct-v0.2, MOSAIC beats uniform in 72 of 76 cells and loses the other four by at most 0.33 points (Tables 2, 3 and 9). The four losses are small and all on LongBench QA: Multi-Doc QA for H2O at B128 (−0.41) and SnapKV at B512 (−0.04), and Single-Doc QA for SnapKV at B256 (−0.74) and AdaKV at B512 (−0.20). Per-dataset



Figure 3: **Learned per-layer allocations, SnapKV and H2O.** (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC’s final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 48 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer’s budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

results, including small dataset-level regressions, are in Appendix I.2.

Winner and NAS-refined. Figure 4 (Long-Bench columns) shows uniform, Stage 3 and Stage 4 at every budget; Table 1 summarizes them per task and method. In the 78 cells that also have a Stage 4 search, the rescaled winner already beats uniform in 67 (75 of 87 over all cells, Table 1); NAS-refined beats uniform in 72 and matches or beats the winner in 54 (Table 1). Refinement adds most on RULER and least on LongBench for SnapKV and H2O (Appendix A). The two stages complement each other across models: on Llama the rescaled winner already captures most of the gain, while on Mistral-7B-Instruct-v0.2 the budget-constrained search does, lifting the cells that beat uniform from 51 of 75 for the rescaled winner to 71 of 76 for NAS-refined (72 of 76 for MOSAIC; Table 9, Figure 5). MOSAIC’s final configuration takes the better of the two, so it holds up on both models.

4.3 Key Findings — RULER

Table 3 lists the results, and Figure 4 (RULER column) shows them per budget. MOSAIC improves on uniform in every RULER cell with a Stage 4 search, for all four methods, by up to +9.7 for SnapKV, +8.3 for AdaKV, +33.3 for H2O and +47.9 for L2Norm; the largest gains come where uniform allocation is weakest.

Gains concentrate on the hardest needle subtasks: for SnapKV and H2O (floor-clean cells), MOSAIC adds +31.9 on niah_multikey_3 and +28.0 on niah_single_3, versus +0.8, +3.5 and +5.6 on CWE, FWE and VT. MOSAIC also recovers most of the needle subtasks that L2Norm’s uniform allocation fails (Appendix I.1).

5 Analysis

5.1 Shared Layer Structure

A few middle layers get more cache. The allocations MOSAIC finds share a common shape (Figure 3). For SnapKV and H2O, five middle layers (10, 14–16 and 20) receive more than an equal

Model	Method	Budget	Code		Multi-Doc QA		Single-Doc QA		Summarization	
			Uniform	MOSAIC	Uniform	MOSAIC	Uniform	MOSAIC	Uniform	MOSAIC
Llama-3-8B	SnapKV	128	54.81	56.09	34.82	35.02	32.97	33.67	21.03	21.15
		256	55.67	57.33	35.60	36.31	35.25	34.51	22.30	22.41
		512	56.73	58.16	35.77	35.73	35.97	36.33	23.44	23.71
		1024	56.02	58.91	35.87	36.26	36.44	37.01	24.85	25.10
	H2O	128	45.96	47.13	32.57	32.16	28.21	29.72	21.88	22.31
		256	49.49	50.59	32.45	32.92	29.96	30.92	23.13	23.33
		512	52.81	54.51	32.98	34.33	32.41	32.80	24.22	24.35
		1024	56.92	57.23	33.54	34.93	33.60	34.69	24.97	25.32
	AdaKV	128	57.30	58.28	34.99	35.96	32.83	34.64	21.72	22.05
		256	58.19	59.62	35.42	36.14	35.08	35.98	22.72	23.00
		512	58.70	60.68	36.14	36.33	36.68	36.49	23.62	24.08
	L2Norm	128	22.54	27.48	9.09	9.23	10.20	10.63	17.73	18.69
		256	23.57	30.69	13.23	18.64	13.91	14.76	19.79	19.98
		512	28.14	35.90	22.06	25.49	18.48	21.13	21.30	21.95
		1024	34.95	44.47	28.72	32.90	24.75	29.21	23.01	23.57
Mistral-7B	SnapKV	128	49.76	49.61	23.86	25.68	29.61	29.50	21.55	–
		256	52.09	53.14	25.62	27.14	31.74	33.53	23.21	–
		512	53.77	55.05	26.47	28.53	33.91	34.39	24.34	–
		1024	55.23	55.38	28.80	29.60	34.99	35.62	25.72	–
	H2O	128	39.59	40.28	21.10	22.29	26.20	27.02	23.50	–
		256	43.09	44.33	23.70	25.16	27.92	29.25	24.55	–
		512	46.43	47.68	25.18	26.43	30.52	31.80	25.14	–
		1024	49.65	50.30	26.58	27.52	32.14	33.50	25.92	–
	AdaKV	128	51.08	51.42	25.27	27.78	30.91	30.57	22.06	–
		256	52.66	53.42	26.66	28.80	33.14	34.12	22.82	–
		512	54.65	55.27	27.62	29.05	34.42	34.21	23.90	–
		1024	55.08	55.75	28.86	30.29	34.97	35.15	24.97	–
	L2Norm	128	22.45	24.89	4.96	5.84	5.03	6.04	18.02	18.81
		256	25.58	28.18	5.73	7.29	7.27	8.88	19.61	20.40
		512	28.68	32.12	7.55	12.96	9.83	14.66	20.89	21.72
		1024	33.22	38.18	11.01	17.45	16.27	23.46	22.48	23.41

Table 2: **LongBench results (full data)**. Uniform allocation vs. MOSAIC’s final configuration (the better of Winner and NAS-refined) at the same average per-layer budget B ; each score is the mean of the category’s datasets (official metrics); the better of the two is in bold. Llama-3-8B: Llama-3-8B-Instruct; Mistral-7B: Mistral-7B-Instruct-v0.2, whose SnapKV, H2O and AdaKV Summarization rows are uniform only (no budget search).

Model	Method		64	128	256	512	1024	1536	2048
Llama-3-8B	SnapKV	Uniform	34.64	51.56	68.06	78.62	85.67	88.62	92.93
		MOSAIC	44.99	58.37	72.28	81.76	93.80	98.35	98.91
	H2O	Uniform	7.23	12.19	20.02	29.64	44.35	53.39	62.82
		MOSAIC	7.60	15.39	19.61	32.61	62.52	86.05	96.14
	AdaKV	Uniform	35.80	50.87	68.02	78.22	86.87	89.75	93.90
		MOSAIC	35.97	57.40	72.10	83.08	93.78	97.99	98.65
	L2Norm	Uniform	–	4.63	17.74	27.17	35.59	42.28	49.70
		MOSAIC	–	13.68	20.83	29.42	67.03	90.19	97.21
Mistral-7B	SnapKV	Uniform	–	40.89	51.74	62.83	77.13	84.66	90.22
		MOSAIC	–	47.38	65.24	71.28	91.06	95.71	96.66
	H2O	Uniform	–	10.14	14.09	19.21	27.68	33.96	41.05
		MOSAIC	–	10.67	14.75	43.01	68.02	85.31	88.63
	AdaKV	Uniform	–	36.33	49.02	60.93	76.71	83.60	87.56
		MOSAIC	–	52.61	60.28	73.94	77.08	88.32	91.71
	L2Norm	Uniform	–	0.64	4.25	11.06	22.86	27.59	30.73
		MOSAIC	–	3.32	10.29	19.43	29.65	43.23	64.58

Table 3: **RULER results (full data, string-match accuracy averaged over 11 subtasks)**. Uniform allocation vs. MOSAIC’s final configuration (the better of Winner and NAS-refined) at the same average per-layer budget; the better is in bold. Llama-3-8B: Llama-3-8B-Instruct, of which SnapKV and H2O at B1024 and above used an earlier 3–4-candidate selection; Mistral-7B: Mistral-7B-Instruct-v0.2.

share of the budget in 65–85% of the 48 allocations, while early (0–9) and late (21–29) layers do so in a median of only 32% and 31% (Figure 3c). The pattern does not come from counting the same shape several times: without the 16 rescaled Winners, which reuse their anchor’s shape at other budgets, layers 10 and 14–16 still receive more than an equal share in 66–84% of allocations (medians 38% and

34% for early and late layers). AdaKV favours the same layers, more weakly (Appendix F).

Shapes depend on the task more than on the method. A winner found for one task does not carry over to another: SnapKV’s Code winner applied to Single-Doc QA, and the reverse, scores below uniform at every budget tested (by 0.33–1.22 points). Across eviction methods on the same task,

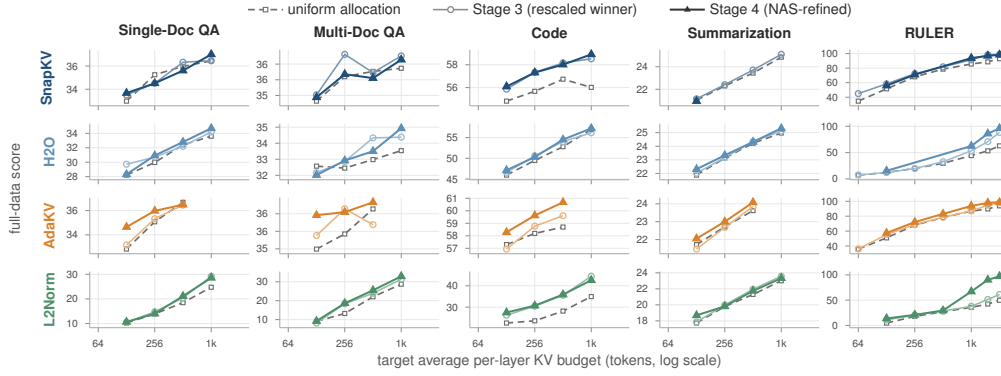


Figure 4: **Uniform vs. Stage 3 and Stage 4 at every target budget (full data).** Rows: eviction method (Llama); columns: task. Stage 3: rescaled Stage 2 winner, no further search; Stage 4: NAS-refined, only at budgets where it ran.

shapes transfer well: H2O’s Code winner used under SnapKV, and SnapKV’s under H2O, beat uniform at every budget (+0.06 to +1.66), although not the receiving method’s own MOSAIC configuration (Appendix C). Each task therefore needs its own search, while a shape found for one eviction method is a good starting point for another.

5.2 Task and Method Patterns

Gains are largest on RULER. Within LongBench they are largest on Code, smaller on the two QA categories and smallest on Summarization, an ordering that holds for every method (Table 1). Across methods, H2O shows the largest RULER gains (+18 to +33 at B1024–B2048). AdaKV, which already allocates budget adaptively across heads within a layer, still gains from reallocation across layers: +4.1 to +8.3 on RULER and up to +1.98 on LongBench. L2Norm shows the largest LongBench gains (Code +4.9 to +9.5), but from a weak baseline: at B128 its uniform Single-Doc QA score is 10.2, against 33.0 for SnapKV.

6 Comparison with EvolKV

We reproduce EvolKV (Yu and Chai, 2025) in its *direct* setting (a search at each budget) and its main *expanded* protocol (one B128 search, rescaled), with SnapKV on Code and Single-Doc QA, all re-evaluated on full data (Table 4).

How the protocols differ. Both rescale one allocation or search at each budget, but find that allocation differently. EvolKV fixes the target budget before searching and returns one allocation; MOSAIC’s Stage 1 treats average budget and task score as separate objectives, and Stage 2 selects one anchor from the Pareto front. EvolKV runs CMA-

Task	Protocol	128	256	512	1024
Code	Uniform	54.81	55.67	56.73	56.02
	EvolKV direct	54.92	–	57.02	57.20
	EvolKV expanded	54.92	55.99	56.82	56.42
	MOSAIC Winner	55.86	57.33	58.16	58.52
	MOSAIC (final)	56.09	57.33	58.16	58.91
Single-Doc QA	Uniform	32.97	35.25	35.97	36.44
	EvolKV direct	33.50	–	36.28	36.40
	EvolKV expanded	33.50	35.43	36.14	36.38
	MOSAIC Winner	33.51	34.51	36.33	36.46
	MOSAIC (final)	33.67	34.51	36.33	37.01

Table 4: **EvolKV vs. MOSAIC (SnapKV, full data).** Direct: an EvolKV search at each budget (none at B256). Expanded: EvolKV’s main protocol, its B128 allocation rescaled. Winner: MOSAIC’s Stage 2 winner, searched once and rescaled. MOSAIC (final): the better of Winner and NAS-refined. Bold: best in each column.

ES on one 8-layer group at a time (earlier groups frozen), starting from uniform, so each search varies 8 layers at one budget (at most $\sim 10^{29}$ allocations). MOSAIC searches all L budgets jointly, about 10^{69} allocations at B128 ($\sim 10^{40} \times$ more per search) and in Stage 1 up to 10^{27} configurations across all budgets (§3.1).

Searching once, MOSAIC’s Winner beats expanded EvolKV in 5 of 6 transferred cells (B256–B1024; Code +1.34 to +2.10, Single-Doc QA -0.92 to $+0.19$). **It also beats direct EvolKV**, a search at each budget, in all 6 cells where both ran (Code +0.94 to +1.32, Single-Doc QA +0.01 to +0.06), with no per-budget search. Search costs and a compute-matched per-budget comparison are in Appendix D.

7 Conclusion

One MOSAIC search per task learns per-layer KV-cache budgets that transfer to every budget, beating uniform allocation in 146 of 154 settings with up to 4 $\times$ less KV cache.

Limitations and Future Work

- **Scope:** we evaluate two dense decoder models; on Mistral-7B-Instruct-v0.2, LongBench Summarization has only Stage 1 for SnapKV, H2O and AdaKV, and we will add Stages 2–4 for it in a future version of this paper.
- **Future work:** we plan to extend MOSAIC to other model families, such as mixture-of-experts models and hybrid architectures that combine attention with convolutional or state-space layers (e.g., LFM and Mamba-based models), and to more recent eviction methods such as KeyDiff (Park et al., 2025) and QUOKA (Jones et al., 2026).

References

- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2024. [Longbench: A bilingual, multi-task benchmark for long context understanding](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3119–3137.
- Han Cai, Chuang Gan, Tianzhe Wang, Zhekai Zhang, and Song Han. 2020. [Once-for-all: Train one network and specialize it for efficient deployment](#). In *International Conference on Learning Representations*.
- Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Yucheng Li, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Junjie Hu, and Wen Xiao. 2025. [PyramidKV: Dynamic KV cache compression based on pyramidal information funneling](#). In *Conference on Language Modeling*.
- Tri Dao. 2024. [Flashattention-2: Faster attention with better parallelism and work partitioning](#). In *The Twelfth International Conference on Learning Representations*.
- Kalyanmoy Deb, Amrit Pratap, Sameer Agarwal, and T. Meyarivan. 2002. [A fast and elitist multiobjective genetic algorithm: NSGA-II](#). *IEEE Transactions on Evolutionary Computation*, 6(2):182–197.
- Alessio Devoto, Yu Zhao, Simone Scardapane, and Pasquale Minervini. 2024. [A simple and effective \$L_2\$ norm-based strategy for KV cache compression](#). In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 18476–18499.
- Thomas Elsken, Jan Hendrik Metzen, and Frank Hutter. 2019. Neural architecture search: A survey. *The Journal of Machine Learning Research*, 20:1–21.
- Yuan Feng, Junlin Lv, Yukun Cao, Xike Xie, and S. Kevin Zhou. 2025. [Ada-KV: Optimizing KV cache eviction by adaptive budget allocation for efficient LLM inference](#). In *Advances in Neural Information Processing Systems*.
- Elias Frantar, Saleh Ashkboos, Torsten Hoeftler, and Dan Alistarh. 2023. [Gptq: Accurate post-training quantization for generative pre-trained transformers](#). In *The Eleventh International Conference on Learning Representations*.
- Yu Fu, Zefan Cai, Abedelkadir Asi, Wayne Xiong, Yue Dong, and Wen Xiao. 2025. [Not all heads matter: A head-level kv cache compression method with integrated retrieval and reasoning](#). In *The Thirteenth International Conference on Learning Representations*.
- Suyu Ge, Yunan Zhang, Liyuan Liu, Minjia Zhang, Jiawei Han, and Jianfeng Gao. 2024. [Model tells you what to discard: Adaptive kv cache compression for llms](#). In *The Twelfth International Conference on Learning Representations*.
- Aaron Grattafiori et al. 2024. [The llama 3 herd of models](#). *arXiv preprint arXiv:2407.21783*.
- Nikolaus Hansen and Andreas Ostermeier. 2001. [Completely derandomized self-adaptation in evolution strategies](#). *Evolutionary Computation*, 9(2):159–195.
- Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Michael W. Mahoney, Yakun Sophia Shao, Kurt Keutzer, and Amir Gholami. 2024. [Kvquant: Towards 10 million context length llm inference with kv cache quantization](#). In *Advances in Neural Information Processing Systems*, volume 37.
- Cheng-Ping Hsieh, Simeng Sun, Samuel Kriman, Shantanu Acharya, Dima Rekesh, Fei Jia, Yang Zhang, and Boris Ginsburg. 2024. [Ruler: What’s the real context size of your long-context language models?](#) In *First Conference on Language Modeling*.
- Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, Chris Bamford, Devendra Singh Chaplot, Diego de las Casas, Florian Bressand, Gianna Lengyel, Guillaume Lample, Lucile Saulnier, L  lio Renard Lavaud, Marie-Anne Lachaux, Pierre Stock, Teven Le Scao, Thibaut Lavril, Thomas Wang, Timoth  e Lacroix, and William El Sayed. 2023. [Mistral 7b](#). *arXiv preprint arXiv:2310.06825*.
- Dalton Jones, Junyoung Park, Matthew Morse, Mingu Lee, Chris Lott, and Harper Langston. 2026. [QUOKA: Query-oriented KV selection for efficient LLM prefill](#). *arXiv preprint arXiv:2602.08722*.
- Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. 2023. [Efficient memory management for large language model serving with pagedattention](#). In *Proceedings of the 29th Symposium on Operating Systems Principles*, pages 611–626.

607	Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. SnapKV: LLM knows what you are looking for before generation. In <i>Advances in Neural Information Processing Systems</i> .	663
608		664
609		
610		
611		
612		
613	Akide Liu, Jing Liu, Zizheng Pan, Yefei He, Gholamreza Haffari, and Bohan Zhuang. 2024a. Minicache: Kv cache compression in depth dimension for large language models. In <i>Advances in Neural Information Processing Systems</i> , volume 37.	
614		
615		
616		
617		
618	Hanxiao Liu, Karen Simonyan, and Yiming Yang. 2019. Darts: Differentiable architecture search. In <i>International Conference on Learning Representations</i> .	
619		
620		
621	Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhao Xu, Anastasios Kyrillidis, and Anshumali Shrivastava. 2023. Scissorhands: Exploiting the persistence of importance hypothesis for llm kv cache compression at test time. In <i>Advances in Neural Information Processing Systems</i> , volume 36.	
622		
623		
624		
625		
626		
627		
628	Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhao Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. 2024b. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. In <i>Proceedings of the 41st International Conference on Machine Learning</i> .	
629		
630		
631		
632		
633	Michael D. McKay, Richard J. Beckman, and William J. Conover. 1979. A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. <i>Technometrics</i> , 21(2):239–245.	
634		
635		
636		
637		
638	Junyoung Park, Dalton Jones, Matthew J. Morse, Raghavv Goel, Mingu Lee, and Chris Lott. 2025. KeyDiff: Key similarity-based KV cache eviction for long-context LLM inference in resource-constrained environments. In <i>Advances in Neural Information Processing Systems</i> .	
639		
640		
641		
642		
643		
644	Ziran Qin, Yuchen Cao, Mingbao Lin, Wen Hu, Shixuan Fan, Ke Cheng, Weiyao Lin, and Jianguo Li. 2025. CAKE: Cascading and adaptive KV cache eviction with layer preferences. In <i>International Conference on Learning Representations</i> .	
645		
646		
647		
648		
649	Rainer Storn and Kenneth Price. 1997. Differential evolution – a simple and efficient heuristic for global optimization over continuous spaces. <i>Journal of Global Optimization</i> , 11(4):341–359.	
650		
651		
652		
653	Jiaming Tang, Yilong Zhao, Kan Zhu, Guangxuan Xiao, Baris Kasikci, and Song Han. 2024. Quest: Query-aware sparsity for efficient long-context llm inference. In <i>Proceedings of the 41st International Conference on Machine Learning</i> .	
654		
655		
656		
657		
658	Louis C. Tiao, Aaron Klein, Matthias Seeger, Edwin V. Bonilla, Cedric Archambeau, and Fabio Ramos. 2021. Bore: Bayesian optimization by density-ratio estimation. In <i>Proceedings of the 38th International Conference on Machine Learning</i> , volume 139 of <i>Proceedings of Machine Learning Research</i> , pages 10289–10300.	665
659		666
660		667
661		668
662		669
	Zihao Wang, Bin Cui, and Shaoduo Gan. 2025. Squeezeattention: 2d management of kv-cache in llm inference via layer-wise optimal budget. In <i>The Thirteenth International Conference on Learning Representations</i> .	
	Guangxuan Xiao, Ji Lin, Mickael Seznec, Hao Wu, Julien Demouth, and Song Han. 2023. Smoothquant: Accurate and efficient post-training quantization for large language models. In <i>Proceedings of the 40th International Conference on Machine Learning</i> .	670
		671
		672
		673
		674
	Guangxuan Xiao, Jiaming Tang, Jingwei Zuo, Junxian Guo, Shang Yang, Haotian Tang, Yao Fu, and Song Han. 2025. Duoattention: Efficient long-context llm inference with retrieval and streaming heads. In <i>The Thirteenth International Conference on Learning Representations</i> .	675
		676
		677
		678
		679
		680
	Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. 2024. Efficient streaming language models with attention sinks. In <i>The Twelfth International Conference on Learning Representations</i> .	681
		682
		683
		684
		685
	Dongjie Yang, XiaoDong Han, Yan Gao, Yao Hu, Shilin Zhang, and Hai Zhao. 2024. Pyramidinfer: Pyramid kv cache compression for high-throughput llm inference. In <i>Findings of the Association for Computational Linguistics: ACL 2024</i> , pages 3258–3270.	686
		687
		688
		689
		690
	Bohan Yu and Yekun Chai. 2025. EvolKV: Evolutionary KV cache compression for LLM inference. In <i>Findings of the Association for Computational Linguistics: EMNLP 2025</i> .	691
		692
		693
		694
	Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023. H ₂ O: Heavy-hitter oracle for efficient generative inference of large language models. In <i>Advances in Neural Information Processing Systems</i> .	695
		696
		697
		698
		699
		700
		701

A Per-Cell Results

LongBench categories: Single-Doc QA (NarrativeQA, Qasper, MultiFieldQA-en), Multi-Doc QA (HotpotQA, 2WikiMQA, MuSiQue), Summarization (GovReport, QMSum, Multi-News) and Code (LCC, RepoBench-P); a category’s score is the mean over its datasets, each scored on all of its 150–500 samples. RULER uses its 11 subtasks (NIAH-Single-1/2/3, NIAH-MultiKey-1/2/3, NIAH-MultiQuery, NIAH-MultiValue, CommonWords Extraction, Frequent-Words Extraction and Variable Tracking) at 4K context, 500 samples each; the RULER score is the mean over the 11 subtasks.

Tables 5 and 6 (Llama-3-8B-Instruct, LongBench and RULER) and Tables 7 and 8 (Mistral-7B-Instruct-v0.2) list every evaluated (task, method, budget) cell on full data. For each task they give three rows: uniform allocation, the rescaled Stage 2 winner (Winner, Stage 3) and NAS-refined, the best configuration of the Stage 4 search. MOSAIC’s final configuration is the better of Winner and NAS-refined. Winner rows also cover budgets where Stage 4 did not run, so they give the full winner transfer of Appendix C. Appendix I breaks these cells down into RULER subtasks and LongBench datasets.

B Calibration Scores vs. Full-Data Scores

MOSAIC scores configurations on a calibration subsample while it searches and on the full data when it selects. Figure 6 shows why both are needed: calibration orders configurations broadly like the full data, but the configuration it ranks first is often not the full-data best, and selection keeps only the first. The Stage 4 searches analysed here use a 10% subsample for all four methods on both benchmarks; Stage 1 uses 30% on LongBench and 10% on RULER.

Stage 4 (Figure 6a, b). All candidates of a Stage 4 cell share one average budget, so their ranking tests calibration directly. Panel (a) gives Kendall’s τ between the calibration and full-data rankings of each cell’s re-evaluated candidates, and panel (b) the regret, the full-data points lost by taking the calibration-best candidate instead of the best re-evaluated one. On LongBench the median τ per method is 0.00 to 0.55, and calibration picks the full-data best in only 4 of 13 cells for SnapKV, 2 of 16 for H2O, 2 of 9 for AdaKV and 10 of 16 for

Task		128	256	512	1024
SnapKV					
Code (430)	Uniform	54.81	55.67	56.73	56.02
	Winner	55.86	57.33	58.16	58.52
	NAS-ref.	56.09	57.31	58.01	58.91
Multi-Doc QA (274)	Uniform	34.82	35.60	35.77	35.87
	Winner	35.02	36.31	35.73	36.26
	NAS-ref.	34.94	35.68	35.55	36.14
Single-Doc QA (506)	Uniform	32.97	35.25	35.97	36.44
	Winner	33.51	34.51	36.33	36.46
	NAS-ref.	33.67	34.51	35.61	37.01
Summarization (732)	Uniform	21.03	22.30	23.44	24.85
	Winner	21.15	22.41	23.71	25.10
	NAS-ref.	20.95	–	–	–
H2O					
Code (2456)	Uniform	45.96	49.49	52.81	56.92
	Winner	46.59	50.59	54.03	56.15
	NAS-ref.	47.13	50.31	54.51	57.23
Multi-Doc QA (448)	Uniform	32.57	32.45	32.98	33.54
	Winner	32.16	32.86	34.33	34.38
	NAS-ref.	32.01	32.92	33.50	34.93
Single-Doc QA (518)	Uniform	28.21	29.96	32.41	33.60
	Winner	29.72	30.70	32.21	34.20
	NAS-ref.	28.32	30.92	32.80	34.69
Summarization (552)	Uniform	21.88	23.13	24.22	24.97
	Winner	22.02	23.17	24.35	25.11
	NAS-ref.	22.31	23.33	24.32	25.32
AdaKV					
Code (482)	Uniform	57.30	58.19	58.70	–
	Winner	56.94	58.77	59.61	–
	NAS-ref.	58.28	59.62	60.68	–
Multi-Doc QA (188)	Uniform	34.99	35.42	36.14	–
	Winner	35.38	36.14	35.69	–
	NAS-ref.	35.96	36.05	36.33	–
Single-Doc QA (464)	Uniform	32.83	35.08	36.68	–
	Winner	33.16	35.33	36.45	–
	NAS-ref.	34.64	35.98	36.49	–
Summarization (548)	Uniform	21.72	22.72	23.62	–
	Winner	21.47	22.68	23.85	–
	NAS-ref.	22.05	23.00	24.08	–
L2Norm					
Code (1410)	Uniform	22.54	23.57	28.14	34.95
	Winner	26.20	30.46	35.55	44.47
	NAS-ref.	27.48	30.69	35.90	42.61
Multi-Doc QA (2236)	Uniform	9.09	13.23	22.06	28.72
	Winner	8.18	18.41	23.91	31.37
	NAS-ref.	9.23	18.64	25.49	32.90
Single-Doc QA (2120)	Uniform	10.20	13.91	18.48	24.75
	Winner	10.52	14.76	20.43	29.21
	NAS-ref.	10.63	14.02	21.13	28.57
Summarization (2518)	Uniform	17.73	19.79	21.30	23.01
	Winner	17.86	19.98	21.95	23.57
	NAS-ref.	18.69	19.82	21.74	23.33

Table 5: **Full-data results for every evaluated LongBench cell, Llama-3-8B-Instruct.** Per task: uniform, the rescaled Stage 2 winner (Winner) and NAS-refined (best of the Stage 4 search); bold: best in each column. The number under each task is the winner’s natural average budget, at which Stage 1 found it. MOSAIC’s final configuration beats uniform in 53 of 57 cells with a Stage 4 search, and the Winner alone in 49 of 60.

Method		64	128	256	512	1024	1536	2048
SnapKV (1756)	Uniform	34.64	51.56	68.06	78.62	85.67	88.62	92.93
	Winner	44.99	58.37	72.28	81.76	91.46	98.35	98.91
	NAS-ref.	–	55.91	71.07	–	93.80	96.86	98.35
H2O (2180)	Uniform	7.23	12.19	20.02	29.64	44.35	53.39	62.82
	Winner	7.60	12.22	19.61	32.61	53.42	70.88	88.31
	NAS-ref.	–	15.39	–	–	62.52	86.05	96.14
AdaKV (2536)	Uniform	35.80	50.87	68.02	78.22	86.87	89.75	93.90
	Winner	35.97	55.32	68.74	78.50	87.09	94.76	98.17
	NAS-ref.	–	57.40	72.10	83.08	93.78	97.99	98.65
L2Norm (2354)	Uniform	–	4.63	17.74	27.17	35.59	42.28	49.70
	Winner	–	10.85	20.50	28.48	37.76	51.33	61.16
	NAS-ref.	–	13.68	20.83	29.42	67.03	90.19	97.21

Table 6: **Full-data results for every evaluated RULER cell, Llama-3-8B-Instruct.** Layout as in Table 5; the number under each method is its winner’s natural average budget. AdaKV at B64: only the rescaled winner was run, compared with Stage 2’s uniform-64 configuration. MOSAIC’s final configuration beats uniform in 21 of 21 cells with a Stage 4 search, and the Winner alone in 26 of 27.

Task		128	256	512	1024
L2Norm					
Code	Uniform	22.45	25.58	28.68	33.22
	Winner	24.40	27.93	31.70	37.87
	NAS-ref.	24.89	28.18	32.12	38.18
Multi-Doc QA	Uniform	4.96	5.73	7.55	11.01
	Winner	5.11	6.10	8.96	16.18
	NAS-ref.	5.84	7.29	12.96	17.45
Single-Doc QA	Uniform	5.03	7.27	9.83	16.27
	Winner	6.04	8.81	14.55	23.15
	NAS-ref.	5.88	8.88	14.66	23.46
Summarization	Uniform	18.02	19.61	20.89	22.48
	Winner	18.16	19.73	21.42	23.00
	NAS-ref.	18.81	20.40	21.72	23.41
SnapKV					
Code	Uniform	49.76	52.09	53.77	55.23
	Winner	48.99	52.52	53.69	55.02
	NAS-ref.	49.61	53.14	55.05	55.38
Multi-Doc QA	Uniform	23.86	25.62	26.47	28.80
	Winner	25.68	27.14	28.53	28.73
	NAS-ref.	25.63	27.11	28.35	29.60
Single-Doc QA	Uniform	29.61	31.74	33.91	34.99
	Winner	29.50	33.34	34.39	35.62
	NAS-ref.	29.13	33.53	34.14	35.42

Task		128	256	512	1024
H2O					
Code	Uniform	39.59	43.09	46.43	49.65
	Winner	40.28	43.98	47.11	49.70
	NAS-ref.	40.25	44.33	47.68	50.30
Multi-Doc QA	Uniform	21.10	23.70	25.18	26.58
	Winner	22.16	25.16	26.42	26.87
	NAS-ref.	22.29	25.07	26.43	27.52
Single-Doc QA	Uniform	26.20	27.92	30.52	32.14
	Winner	27.02	29.25	31.79	32.77
	NAS-ref.	26.15	28.96	31.80	33.50
AdaKV					
Code	Uniform	51.08	52.66	54.65	55.08
	Winner	46.64	51.19	53.90	54.89
	NAS-ref.	51.42	53.42	55.27	55.75
Multi-Doc QA	Uniform	25.27	26.66	27.62	28.86
	Winner	27.78	28.57	28.66	29.22
	NAS-ref.	27.43	28.80	29.05	30.29
Single-Doc QA	Uniform	30.91	33.14	34.42	34.97
	Winner	28.89	32.23	33.82	35.15
	NAS-ref.	30.57	34.12	34.21	35.07

Table 7: **Full-data results for every evaluated LongBench cell, Mistral-7B-Instruct-v0.2.** Layout as in Table 5. SnapKV, H2O and AdaKV were searched with 30% calibration and have no Summarization cells (only its Stage 1 search ran); H2O truncates prompts at 7.5K tokens, SnapKV and AdaKV at 31.5K. MOSAIC’s final configuration beats uniform in 48 of 52 cells, and the Winner alone in 40 of 52.

L2Norm; on RULER it picks the full-data best in 20 of 21 cells (Table 10).

Stage 1 fronts (Figure 6c, d). Panels (c) and (d) compare each Stage 1 Pareto front’s calibration scores with its Stage 2 full-data re-evaluation. On a front both scores rise with budget, so the front’s overall order agrees (τ 0.78 to 1.00; in 15 of 17

fronts τ equals that between average budget and full-data score). Stage 2, however, keeps only the top configuration. The calibration-top configuration is the most expensive on the front in 15 of 17 fronts, and full data prefers a cheaper one in 6 of 17 (by up to 0.73 points; none of L2Norm’s 4 LongBench fronts). For AdaKV, the Single- and

Method		128	256	512	1024	1536	2048
L2Norm	Uniform	0.64	4.25	11.06	22.86	27.59	30.73
	Winner	2.74	8.78	19.43	25.62	37.30	62.71
	NAS-ref.	3.32	10.29	17.90	29.65	43.23	64.58
SnapKV	Uniform	40.89	51.74	62.83	77.13	84.66	90.22
	Winner	38.69	50.64	62.27	77.15	84.40	88.07
	NAS-ref.	47.38	65.24	71.28	91.06	95.71	96.66
H2O	Uniform	10.14	14.09	19.21	27.68	33.96	41.05
	Winner	9.61	13.67	18.57	37.13	54.44	82.93
	NAS-ref.	10.67	14.75	43.01	68.02	85.31	88.63
AdaKV	Uniform	36.33	49.02	60.93	76.71	83.60	87.56
	Winner	34.23	47.67	61.48	—	83.12	86.43
	NAS-ref.	52.61	60.28	73.94	77.08	88.32	91.71

Table 8: **Full-data results for every evaluated RULER cell, Mistral-7B-Instruct-v0.2.** Layout as in Table 6. MOSAIC’s final configuration beats uniform in 24 of 24 cells, and the Winner alone in 11 of 23.

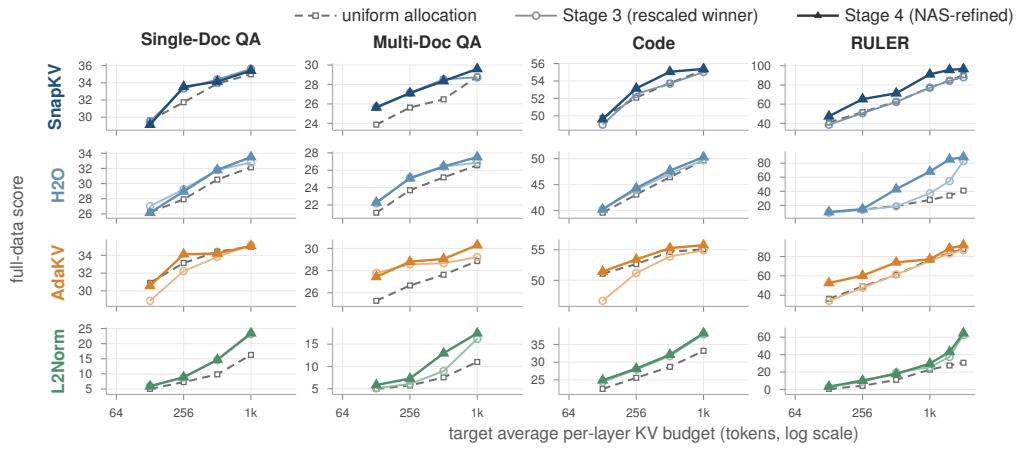


Figure 5: **Uniform vs. Stage 3 and Stage 4 at every target budget, Mistral-7B-Instruct-v0.2** (full data), drawn as Figure 4. Rows: eviction method; columns: task. Stage 3: rescaled Stage 2 winner, no further search; Stage 4: NAS-refined, only at budgets where it ran.

Multi-Doc QA fronts are shown. Re-scoring a front costs only its 4 to 19 configurations.

Calibration ratio: a case study. Table 11 rescores the 24 configurations of one SnapKV Stage 4 search (Code, B512) on a 30% subsample and compares both subsamples with the full data. In this cell the larger subsample reduces the optimism (+1.37 against +5.10 points), ranks the configurations closer to the full data (τ 0.75 against 0.51), and picks the full-data best, which the 10% subsample does not. This is a single cell, chosen because its guided proposal scored below uniform, so it illustrates rather than establishes the effect; a 30% subsample also costs about three times as much per evaluation. A larger subsample reduces the noise but does not remove it: on the Stage 1 fronts, which use 30% on LongBench, full data prefers a different configuration from the calibration-top one in 4 of 15 SnapKV, H2O and L2Norm fronts (by up to 0.73

points; Figure 6d). MOSAIC therefore re-evaluates its candidates on full data at either ratio.

C Shape Transfer Across Tasks and Methods

Figure 7 applies a Stage 2 winner shape to a different task or eviction method, rescaled with the same mapping as Stage 3, and compares it with the receiving cell’s uniform allocation, its own rescaled Winner and its final configuration. The tests cover SnapKV and H2O on Code and Single-Doc QA at B128, B512 and B1024.

Across tasks, shapes do not transfer. SnapKV’s Code shape applied to Single-Doc QA, and its Single-Doc QA shape applied to Code, score below uniform at every budget (by 0.33 to 1.08 and 0.63 to 1.22 points), while each task’s own Winner beats uniform in the same six cells. The layers that need more cache differ between the

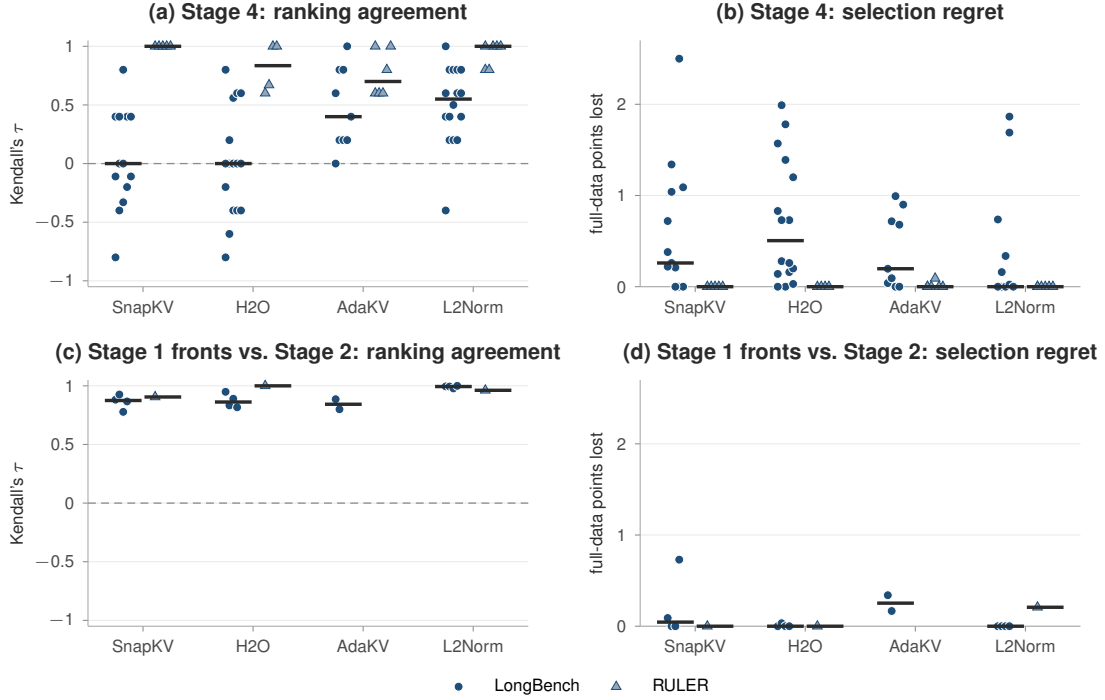


Figure 6: **Calibration vs. full-data scores, Llama-3-8B-Instruct.** Circles: LongBench; triangles: RULER; bars: medians. Top: Stage 4 (10% subset on both benchmarks), one marker per evaluated (task, budget) cell, whose candidates share one budget. Bottom: Stage 1 Pareto fronts (30% on LongBench, 10% on RULER) against their Stage 2 full-data re-evaluation, one marker per front (AdaKV: Single- and Multi-Doc QA). (a, c) Kendall’s τ between the calibration and full-data rankings. On a front both scores rise with budget, so (c) is high and in 15 of 17 fronts equals τ between average budget and full-data score: it reflects the budget trend, not whether calibration finds the top configuration. (b, d) Regret: full-data points lost by taking the calibration-top configuration instead of the best re-evaluated one. In (d), full data prefers a cheaper configuration than the calibration-top one in 6 of 17 fronts (up to 0.73 points). “Best” refers to the re-evaluated pool only, and a zero median regret does not mean calibration always picks correctly (Table 10 gives the Stage 4 counts).

two tasks, so a shape found for one task is worse than uniform allocation on the other.

Across eviction methods, shapes do transfer. On Code, H2O’s shape under SnapKV and SnapKV’s shape under H2O beat uniform in all six cells (+0.06 to +1.66). They stay below the receiving method’s own final configuration in every cell (by 0.18 to 1.23 points) and below its own Winner in 4 of 6; SnapKV’s shape under H2O beats H2O’s own Winner at B128 and B1024 (+0.16 and +0.90). Which layers need more cache therefore depends mainly on the task, so a shape found with one eviction method is a useful starting point for another, while a search with the receiving method still adds up to 1.23 points.

D Search Cost

Table 15 compares one budget-constrained search per cell with our EvolKV reproduction (SnapKV, LongBench), on the same A100-40GB machine

and evaluation harness. EvolKV runs 60 generations $\times$ population 10, 600 evaluations on 30 calibration samples each. MOSAIC’s Stage 4 runs at most 150 evaluations on a 10% calibration subset (55–100 samples): we keep each run’s evaluations ≤ 150 in the order they were run, and add the full-data re-evaluation of the three NAS-refined candidates. Both searches process a sample equally fast (about 2.5 s on Code and 1.0 s on Single-Doc QA), so cost is evaluations times samples, and MOSAIC spends it on fewer evaluations of larger samples. It costs no more than EvolKV in any cell (43.8 GPU-hours against 53.7 in total) and beats it in 5 of 6 cells (+0.17 to +1.71); it loses Single-Doc QA B512 by 0.67 points.

MOSAIC has four stages. Stages 1–2 run once per (task, method) and produce the Winner; Stages 3–4 rescale it to each budget and refine it there. Users can stop after Stages 1–2 and deploy the rescaled Winner, which beats direct EvolKV in all

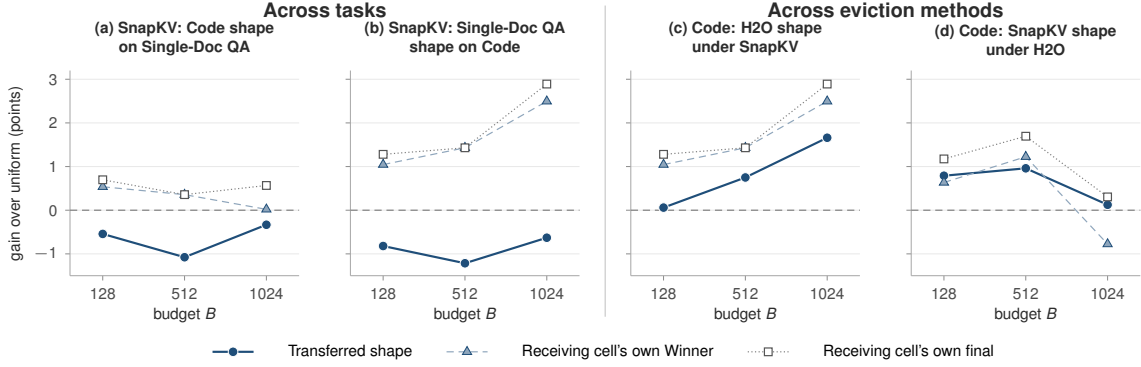


Figure 7: **Transferring winner shapes across tasks (a, b) and eviction methods (c, d), LongBench, full data.** Each point is a gain over the receiving cell’s uniform allocation at the same budget (dashed line): the transferred shape, rescaled as in Stage 3; the receiving cell’s own rescaled Winner; and its own final configuration (the better of Winner and NAS-refined).

Task	Method	Stage 3 (Winner)		Stage 4 (NAS-refined)		MOSAIC (best of both)	
		Wins	Δ	Wins	Δ	Wins	Δ
Code	H2O	4/4	+0.58	4/4	+0.95	4/4	+0.96
Code	SnapKV	1/4	-0.16	3/4	+0.59	3/4	+0.59
MultiDoc	H2O	4/4	+1.01	4/4	+1.19	4/4	+1.21
MultiDoc	SnapKV	3/4	+1.33	4/4	+1.48	4/4	+1.55
SingleDoc	H2O	4/4	+1.01	3/4	+0.91	4/4	+1.19
SingleDoc	SnapKV	3/4	+0.65	3/4	+0.49	3/4	+0.70
RULER	H2O	3/6	+11.71	6/6	+27.38	6/6	+27.38
RULER	SnapKV	1/6	-1.04	6/6	+9.98	6/6	+9.98
Code	AdaKV	0/4	-1.71	4/4	+0.60	4/4	+0.60
MultiDoc	AdaKV	4/4	+1.45	4/4	+1.79	4/4	+1.88
SingleDoc	AdaKV	1/4	-0.83	2/4	+0.14	2/4	+0.16
RULER	AdaKV	1/5	-0.90	6/6	+8.30	6/6	+8.30
Code	L2Norm	4/4	+2.99	4/4	+3.36	4/4	+3.36
MultiDoc	L2Norm	4/4	+1.78	4/4	+3.57	4/4	+3.57
SingleDoc	L2Norm	4/4	+3.54	4/4	+3.62	4/4	+3.66
Summ	L2Norm	4/4	+0.33	4/4	+0.83	4/4	+0.83
RULER	L2Norm	6/6	+9.91	6/6	+11.98	6/6	+12.23
Total		51/75		71/76		72/76	

Table 9: **One search, many budgets, Mistral-7B-Instruct-v0.2** (Table 1 for Llama-3-8B-Instruct; columns as there). SnapKV, H2O and AdaKV have no LongBench Summarization cells.

Bench.	Method	Cells	Gap	τ	Picks	Regret
LongBench	H2O	16	+3.77	-0.00	2/16	0.71
LongBench	SnapKV	13	+5.23	+0.03	4/13	0.60
RULER	H2O	4	+0.09	+0.82	4/4	0.00
RULER	SnapKV	5	-0.16	+1.00	5/5	0.00
LongBench	AdaKV	9	+1.14	+0.47	2/9	0.40
RULER	AdaKV	6	-0.20	+0.77	5/6	0.02
LongBench	L2Norm	16	+2.58	+0.49	10/16	0.30
RULER	L2Norm	6	+0.13	+0.93	6/6	0.00

Table 10: **Stage 4 calibration (10% subset) vs. full-data scores, summary.** Cells: (task, budget) pairs. Gap: mean calibration minus full-data score. τ : mean Kendall correlation between the two rankings of each cell’s re-evaluated configurations. Picks: cells where the calibration-best candidate is also full-data best. Regret: mean full-data points lost by trusting calibration.

Subsample	Gap	τ	Picks best	Regret
10%	+5.10	+0.51	no	2.87
30%	+1.37	+0.75	yes	0.00

Table 11: **Calibration ratio, one cell** (SnapKV, Code B512, 24 configurations, full data as reference). Gap: mean calibration minus full-data score.

Category	Method	GPU-h
SingleDoc	SnapKV	9.0
SingleDoc	H2O	26.0
MultiDoc	H2O	29.4
RULER	SnapKV	22.9
RULER	H2O	39.7

Table 12: **MOSAIC Stage 1 cost**

6 cells (Section 6), or also run Stage 4 at the budgets they need. Stage 2 re-evaluates the Stage 1 front on full data: 3.6 GPU-hours on Single-Doc QA (20 configurations) and 8.0 on Code (11 configurations). Table 12 gives the cost of the 180-evaluation Stage 1 for each task and method. With Stages 1–2 added, running all four stages costs more than EvolKV’s direct searches at these three budgets (at least 43.1 vs. 38.3 GPU-hours on Code, 21.2 vs. 15.4 on Single-Doc QA).

E Choice of the Per-Layer Floor

SnapKV and H2O always retain an 8-token recent window, so a layer with budget b selects only $b - 8$ tokens by attention. At a floor of 16, only 8 tokens are chosen from contexts of thousands. Table 13 shows how often MOSAIC’s final configuration in each cell places layers exactly at the floor. At B128, 8 to 24 of the 32 layers sit at the floor, so the floor matters most at low budgets, which is where the controlled comparison below finds its differences.

Task	Method	128	256	512	1024
Code	SnapKV	16	15	0	1
	H2O	18	8	2	1
MultiDoc	SnapKV	24	0	0	0
	H2O	17	4	0	17
SingleDoc	SnapKV	8	3	0	1
	H2O	9	3	1	0
Summ	SnapKV	8	6	0	0
	H2O	13	6	0	0
RULER	SnapKV	14	11	0	10
	H2O	19	0	0	21

Table 13: **Layers at the per-layer floor (out of 32) in MOSAIC’s final configuration of each cell.**

Controlled comparison. Table 14 decodes the same rescaled SnapKV winner under both floors and evaluates both on full data. Floor 64 is better in three cells, all at B128 (by 0.37–1.10 points), the two are tied in two cells, and floor 16 is better in one (by 0.26). The differences concentrate at the lowest budget, where the floor binds most often.

Cell	Uniform	Floor 64	Floor 16	Δ (16–64)
Code B128	54.81	55.86	54.76	–1.10
Code B256	55.67	57.33	57.30	–0.02
SingleDoc B128	32.97	33.51	33.14	–0.37
SingleDoc B256	35.25	34.51	34.77	+0.26
MultiDoc B128	34.82	35.02	34.32	–0.70
MultiDoc B256	35.60	36.31	36.31	+0.00

Table 14: **Same rescaled winner under floor 64 and floor 16** (SnapKV, LongBench, full data). Negative Δ (bold): floor 64 is better.

F AdaKV and L2Norm Per-Layer Allocations

Figure 8 is the AdaKV counterpart of Figure 3, drawn in the same way. AdaKV shows a weaker version of the middle-layer preference: layer 10 receives more than its uniform share in 95% of the 19 AdaKV allocations and layers 13–16 in 68–89%,

but over the whole middle block (layers 10 and 14–20) the share is 55%, against 65% for SnapKV and H2O. Figure 9 does the same for L2Norm. Its allocations favour layers 14–18, which receive more than their uniform share in 56–70% of the 27 L2Norm allocations, and layer 27 (70%); over the whole middle block the share is 51%.

G Further Related Work

Complementary cache compression. Other techniques shrink the KV cache along axes that per-layer allocation leaves untouched and can be combined with it: Quest (Tang et al., 2024) keeps the full cache but loads only the pages relevant to each query, KIVI (Liu et al., 2024b) and KVQuant (Hooper et al., 2024) quantize keys and values to a few bits, and MiniCache (Liu et al., 2024a) merges the caches of adjacent layers.

Searching once for many budgets. Once-for-All (Cai et al., 2020) trains one network and specializes it to many hardware budgets; MOSAIC similarly searches once and rescales the result to many cache budgets. Gradient-based NAS such as DARTS (Liu et al., 2019) does not apply here, because the task score is not differentiable in the per-layer token budgets.

H Reproducibility Details

Algorithm 1 summarizes the MOSAIC pipeline for one (task, eviction method); the paragraphs below give its settings.

Budget decoding. Stage 1 maps each coordinate $x_i \in [0, 1]$ of a configuration to a discrete per-layer budget by equal-width bins over a grid G : seven levels from 64 to 4096 on RULER and in some LongBench runs, and five levels from 64 to 1024 in the other LongBench runs. The grid limits only Stage 1: Stages 3 and 4 decode continuously between the floor and 4096, scaling the weights x_i to the total $T = 32B$, clamping layers at the cap and the floor and redistributing the remainder in proportion to x_i , then rounding so that $\sum_i b_i = T$ exactly. *Search-space size.* The Stage 1 grid holds $|G|^{32}$ configurations ($5^{32} \approx 2.3 \times 10^{22}$ or $7^{32} \approx 1.1 \times 10^{27}$) across all average budgets. Stage 4 searches integer allocations in $[64, 4096]^{32}$ at a fixed budget B : about 7×10^{68} at B128, and 10^{84} , 10^{95} , 10^{105} and 10^{111} at B256, B512, B1024 and B2048. One EvolKV search varies a single 8-layer group with the others fixed, so even ignoring its budget penalty it covers at most $4033^8 \approx 7 \times$

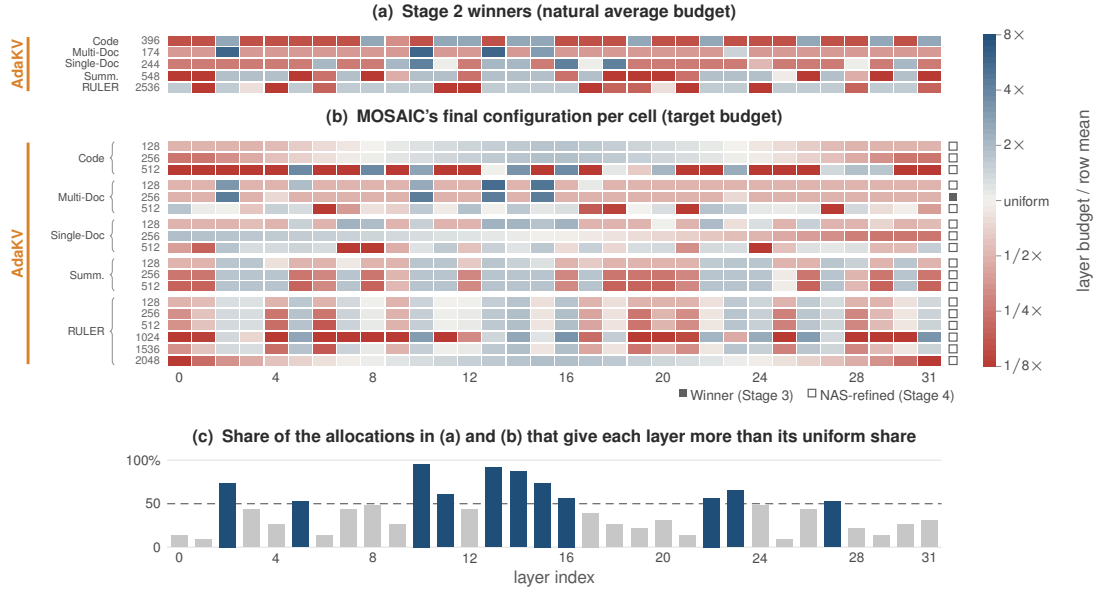


Figure 8: **AdaKV per-layer allocations**, drawn as Figure 3. (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC's final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 19 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer's budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

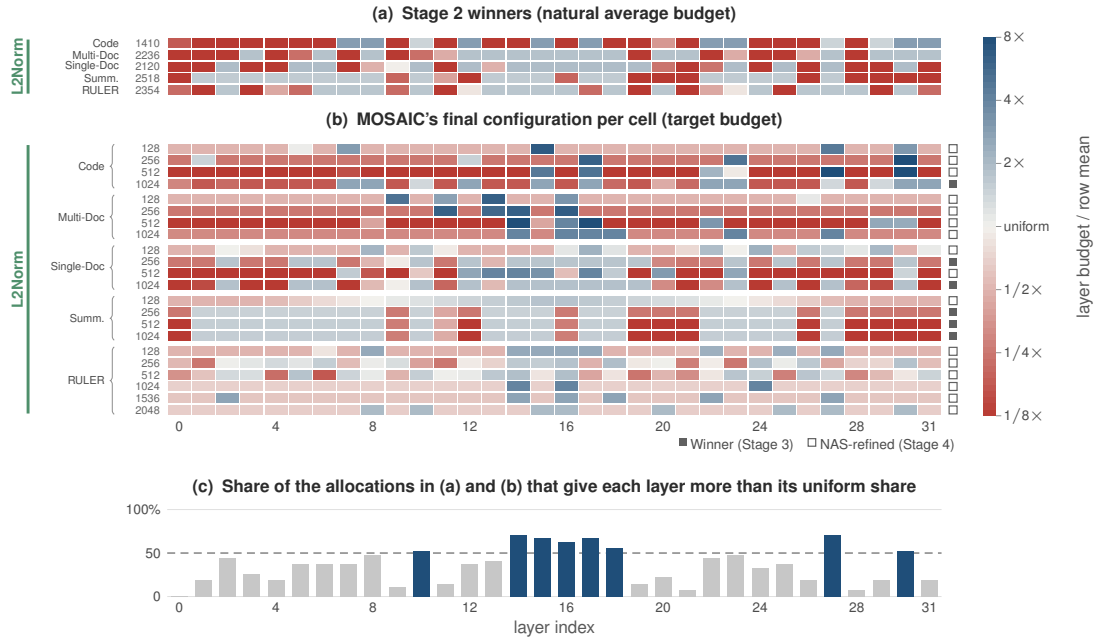


Figure 9: **L2Norm per-layer allocations**, drawn as Figure 3. (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC's final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 27 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer's budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

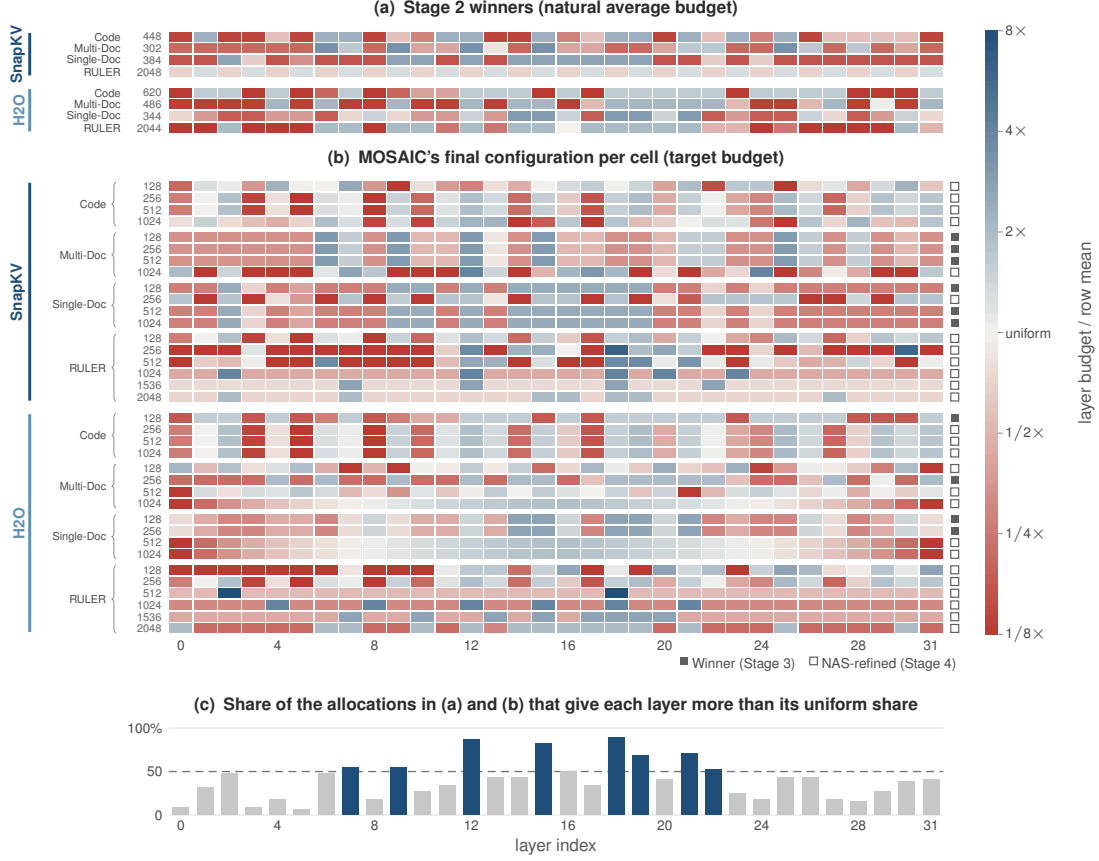


Figure 10: **SnapKV and H2O per-layer allocations, Mistral-7B-Instruct-v0.2**, drawn as Figure 3. (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC's final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 44 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer's budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

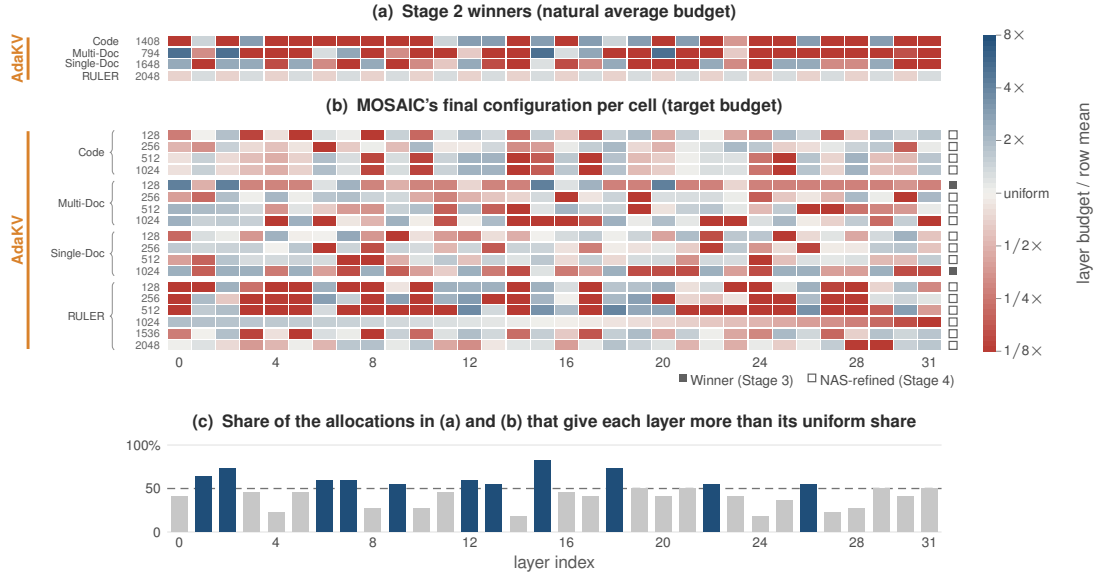


Figure 11: **AdaKV per-layer allocations, Mistral-7B-Instruct-v0.2**, drawn as Figure 3. (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC's final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 22 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer's budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

Category	Budget	EvolKV, direct				MOSAIC Stage 4, NAS-refined				Δ
		Evals	Samples	GPU-h	Score	Evals	Samples	GPU-h	Score	
Code	128	601	30	12.7	54.92	150	100	12.6	55.73	+0.81
Code	512	601	30	12.9	57.02	148	100	12.0	58.01	+0.99
Code	1024	601	30	12.7	57.20	125	100	10.6	58.91	+1.71
SingleDoc	128	601	30	5.2	33.50	150	55	2.8	33.67	+0.17
SingleDoc	512	601	30	5.1	36.28	150	55	3.0	35.61	-0.67
SingleDoc	1024	601	30	5.1	36.40	150	55	2.8	37.01	+0.61
Total				53.7				43.8		5/6

Table 15: **Per-budget search cost and score, EvolKV direct vs. MOSAIC Stage 4** (SnapKV, LongBench, full-data scores, A100). EvolKV: 600 evaluations plus one final evaluation. MOSAIC: ≤ 150 evaluations of each Stage 4 run; its GPU-hours include the full-data re-evaluation of three candidates but exclude the one-time Stages 1–2. Δ : NAS-refined minus EvolKV; losses in bold.

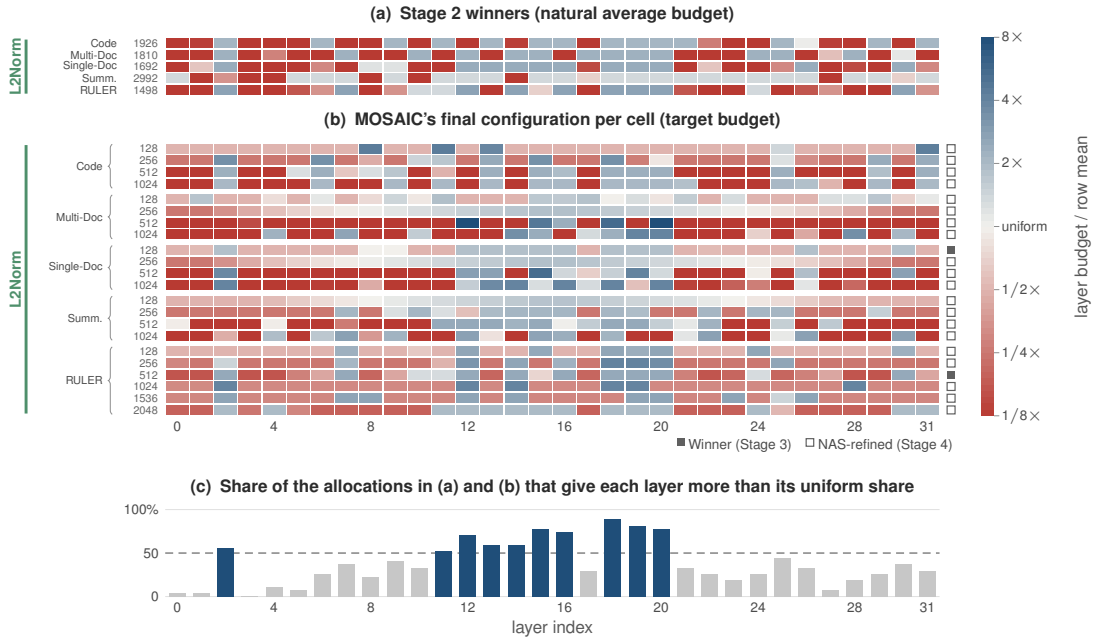


Figure 12: **L2Norm per-layer allocations, Mistral-7B-Instruct-v0.2**, drawn as Figure 3. (a) Stage 2 winners at their natural average budget (left). (b) MOSAIC’s final configuration for every cell (the better of Winner and NAS-refined), with the target budget on the left and a marker on the right for which of the two it is. (c) For each layer, the share of the 27 allocations in (a) and (b) that give it more than its uniform share; navy bars exceed 50%. Colour: each layer’s budget divided by the row mean, on a $\log_2$ scale (blue: more than the uniform share; red: less).

10^{28} allocations, about 10^{40} times fewer than one MOSAIC search at B128. This compares single searches: the allocations of both methods lie in the same space $[64, 4096]^{32}$.

Rescaling the Stage 2 winner. A winner with per-layer budgets w_i is mapped to $x_i = 0.05 + 0.9 w_i / \max_j w_j$ and decoded at each target B as above; this keeps the order of the layers and the shape of the allocation but compresses the ratios between layers. The same mapping seeds the winner into the Stage 4 search.

Per-layer floor and cap. The per-layer cap is 4096. Stage 1’s grid maximum is 1024 for most

LongBench runs and 4096 for RULER; Stages 3 and 4 decode continuously between the floor (64 or 16) and the cap (4096). Table 13 reports how often the floor binds.

Stopping rules. Searches were capped at 180 evaluations in Stage 1, 150 evaluations in Stage 4 and were stopped earlier, by hand or when the best calibration score stopped improving, so run lengths vary across methods and cells.

Candidate provenance. Stage 2 re-evaluates configurations from the Stage 1 Pareto front on full data and takes the best non-uniform one as the Winner. Stage 4 re-evaluates on full data uni-

Algorithm 1 MOSAIC for one (task, eviction method).

Require: calibration subset D_{cal} , full data D_{full} , target budgets $\mathcal{B}$, Stage 1 grid G , evaluation cap N

- 1: *Stage 1:* $A \leftarrow$ uniform allocation at every level of $G \cup$ Latin Hypercube points $\triangleright 64$ points
- 2: $A \leftarrow \text{GUIDEDSEARCH}(A, \text{minimize average budget, maximize score on } D_{\text{cal}})$
- 3: $P \leftarrow$ Pareto front of A
- 4: *Stage 2:* $w \leftarrow$ the non-uniform $p \in P$ with the highest score on D_{full}
- 5: **for** each target budget $B \in \mathcal{B}$ **do**
- 6: *Stage 3:* $b_W \leftarrow \text{RESCALE}(w, B)$ $\triangleright$ Winner
- 7: *Stage 4 (optional):* $A_B \leftarrow \{\text{uniform, 5 heuristic shapes, } b_W, 57 \text{ LHS points}\}$, all with mean budget B
- 8: $A_B \leftarrow \text{GUIDEDSEARCH}(A_B, \text{maximize score on } D_{\text{cal}})$
- 9: $C \leftarrow$ the calibration-best candidates of $A_B \setminus \{\text{uniform, } b_W\}$
- 10: $b_N \leftarrow$ the $c \in C$ with the highest score on D_{full} $\triangleright$ NAS-refined
- 11: final configuration at $B \leftarrow$ the better of b_W and b_N on D_{full}
- 12: **end for**
- 13: **procedure** GUIDEDSEARCH(A , objectives)
- 14: **repeat**
- 15: rank A by non-dominated sorting on the objectives
- 16: train an MLP classifier to separate rank-1 configurations from the rest
- 17: $\mathbf{x}^* \leftarrow \arg \max_{\mathbf{x}} P(\text{rank } 1 \mid \mathbf{x})$, by differential evolution
- 18: evaluate $\mathbf{x}^*$ on D_{cal} ; $A \leftarrow A \cup \{\mathbf{x}^*\}$
- 19: **until** the search stops improving or $|A| = N$
- 20: **return** A
- 21: **end procedure**

form, the Winner and the calibration-best of the search’s other candidates (heuristic shapes, LHS points and guided proposals). NAS-refined is the best of these other candidates, so it excludes uniform and the Winner and can score below the Winner; MOSAIC’s final configuration is the better of the two.

Stage 4 initial design. Each Stage 4 search starts from 64 configurations at the target budget: uniform, five heuristic shapes (ascending and descending ramps, a middle-heavy triangle, its inverse, and alternating low/high layers), the Winner, and 57 Latin hypercube points, which serve as a random-allocation control. Every later evaluation is one configuration proposed by GUIDEDSEARCH.

EvolKV baseline (expanded). To reproduce EvolKV’s main protocol, the allocation k_i that our EvolKV run finds at B128 is expanded to each target B with EvolKV’s budget completion rule (Yu and Chai, 2025), $b_i = k_i T / \sum_j k_j$, rounded so that the mean is exactly B and clipped to $[64, 4096]$

(neither bound binds for B256–B1024).

Hardware. Experiments ran on NVIDIA A100 GPUs, with FlashAttention-2 (Dao, 2024) as the attention implementation for every method.

I Subtask and Dataset Breakdown

This section breaks the per-cell results of Appendix A into RULER subtasks and LongBench datasets, comparing uniform allocation with MOSAIC’s final configuration (the better of Winner and NAS-refined) in every cell; bold marks the better of the two in each column.

I.1 RULER Subtask Breakdown

Tables 16 (Llama-3-8B-Instruct) and 17 (Mistral-7B-Instruct-v0.2) break each RULER cell into its 11 subtasks (S1–3: niah_single; MK1–3: niah_multikey; MQ: multiquery; MV: multivalue; CWE/FWE: common/frequent-word extraction; VT: variable tracking; NIAH: mean of the eight needle subtasks; All: mean of all 11), comparing uniform with MOSAIC’s final configuration; Figures 13 and 14 show the same gains as heatmaps. Averaged over floor-clean cells (floor 64, plus B1536/B2048), NAS gains concentrate on the hardest needle subtasks: MK3 +31.9, S3 +28.0, S1 +23.8, versus CWE +0.8, FWE +3.5 and VT +5.6. The floor-16 H2O B1024 configuration trades CWE accuracy (98.3→21.5) for needle retrieval.

For AdaKV the pattern matches SnapKV and H2O: the largest gains are on niah_multikey_3 at B1024 and above.

I.2 LongBench Dataset Breakdown

LongBench scores are category means over two or three datasets. Tables 18 (Llama-3-8B-Instruct) and 19 (Mistral-7B-Instruct-v0.2) break each LongBench cell into its datasets, comparing uniform with MOSAIC’s final configuration (the better of Winner and NAS-refined on the category mean); Figures 15 and 16 show the per-dataset gains as heatmaps. On Llama, the final configuration falls below uniform on at least one dataset in 23 of 60 cells, and in 19 of them the category mean is still above uniform; on Mistral this happens in 12 of 52 cells.

Budget		S1	S2	S3	MK1	MK2	MK3	MQ	MV	CWE	FWE	VT	NIAH	All
SnapKV														
64	Uniform	99.8	93.4	0.0	91.2	48.0	0.0	7.9	17.0	1.8	1.3	20.6	44.7	34.6
	MOSAIC	100.0	97.2	0.0	96.6	76.4	0.0	23.4	36.0	2.9	34.9	27.4	53.7	45.0
128	Uniform	100.0	98.6	0.2	98.0	72.8	0.0	36.2	53.1	8.2	49.5	50.6	57.4	51.6
	MOSAIC	100.0	98.4	0.2	98.4	91.4	0.0	57.8	64.0	9.3	56.4	66.1	63.8	58.4
256	Uniform	100.0	99.8	1.4	99.4	93.0	0.0	82.5	84.5	30.8	68.0	89.3	70.1	68.1
	MOSAIC	100.0	99.6	10.2	99.4	98.0	0.0	90.2	92.3	38.0	72.5	94.7	73.7	72.3
512	Uniform	100.0	100.0	23.4	99.4	99.6	0.0	97.8	96.5	73.4	78.7	96.2	77.1	78.6
	MOSAIC	100.0	100.0	49.6	99.4	99.6	0.6	98.5	97.5	77.1	81.3	95.8	80.6	81.8
1024	Uniform	100.0	100.0	64.8	99.4	99.8	0.6	99.8	98.8	96.1	84.7	98.4	82.9	85.7
	MOSAIC	100.0	100.0	94.2	99.2	99.6	67.8	99.3	97.8	88.2	89.8	95.9	94.7	93.8
1536	Uniform	100.0	100.0	83.8	99.4	100.0	6.8	100.0	98.8	99.0	88.1	98.9	86.1	88.6
	MOSAIC	100.0	100.0	99.4	99.4	99.8	97.2	99.8	98.8	98.3	90.9	98.4	99.3	98.4
2048	Uniform	100.0	100.0	92.8	99.4	100.0	43.0	99.9	98.9	99.6	89.4	99.2	91.8	92.9
	MOSAIC	100.0	100.0	100.0	99.6	100.0	98.6	99.9	98.8	99.6	92.6	98.9	99.6	98.9
H2O														
64	Uniform	0.2	0.6	0.0	0.4	1.4	0.0	0.1	0.3	11.3	58.9	6.2	0.4	7.2
	MOSAIC	1.0	0.8	0.0	1.6	1.0	0.0	0.8	0.6	8.5	67.2	2.2	0.7	7.6
128	Uniform	2.8	2.8	0.0	4.8	4.6	0.0	1.8	1.8	30.3	78.9	6.4	2.3	12.2
	MOSAIC	13.6	2.8	0.0	7.4	10.8	0.0	3.7	5.9	29.6	86.1	9.4	5.5	15.4
256	Uniform	9.2	11.6	0.0	13.4	16.2	0.2	6.6	7.1	58.1	88.2	9.6	8.0	20.0
	MOSAIC	2.4	6.8	0.0	14.6	24.0	0.2	6.7	6.6	54.0	89.9	10.5	7.7	19.6
512	Uniform	6.4	25.0	0.0	24.2	35.2	1.4	14.4	14.3	86.1	94.3	24.6	15.1	29.6
	MOSAIC	8.4	14.4	0.0	27.6	48.6	1.6	12.8	16.6	85.3	93.4	50.1	16.2	32.6
1024	Uniform	10.2	44.2	0.0	47.0	57.6	7.6	24.8	30.4	98.3	96.3	71.6	27.7	44.4
	MOSAIC	96.6	92.0	63.0	80.2	75.8	1.0	62.4	63.7	21.5	77.5	54.1	66.8	62.5
1536	Uniform	16.2	58.8	0.0	61.0	71.8	24.6	35.6	44.4	99.3	95.1	80.4	39.0	53.4
	MOSAIC	96.2	96.8	65.0	98.0	99.4	51.6	80.8	77.6	98.5	95.7	86.9	83.2	86.0
2048	Uniform	24.0	79.4	0.0	70.4	83.2	48.6	47.6	58.0	99.6	95.1	85.0	51.4	62.8
	MOSAIC	100.0	100.0	99.4	99.6	99.0	98.6	90.7	82.3	98.8	94.7	94.5	96.2	96.1
AdaKV														
64	Uniform	99.6	92.0	0.0	91.6	51.6	0.0	8.2	26.7	4.0	0.7	19.4	46.2	35.8
	MOSAIC	99.6	92.4	0.0	91.4	37.8	0.0	13.9	31.4	4.9	0.2	24.0	45.8	36.0
128	Uniform	99.8	97.8	0.0	97.4	68.2	0.0	44.2	65.8	15.3	16.8	54.2	59.1	50.9
	MOSAIC	100.0	98.6	0.6	98.4	80.0	0.0	59.0	75.5	14.7	43.2	61.5	64.0	57.4
256	Uniform	100.0	99.8	4.2	98.8	84.8	0.0	89.5	90.4	28.9	68.1	83.7	70.9	68.0
	MOSAIC	100.0	100.0	10.8	99.4	94.8	0.0	93.5	96.0	39.7	70.4	88.4	74.3	72.1
512	Uniform	100.0	100.0	32.6	99.6	97.4	0.0	98.5	97.3	59.0	81.9	94.2	78.2	78.2
	MOSAIC	100.0	100.0	58.8	99.4	98.2	0.0	99.6	97.8	81.0	82.4	96.6	81.7	83.1
1024	Uniform	100.0	100.0	77.4	99.6	99.2	1.6	99.8	98.0	93.6	87.3	98.9	84.5	86.9
	MOSAIC	100.0	100.0	96.6	99.6	99.4	74.6	97.0	96.5	86.4	86.9	94.6	95.5	93.8
1536	Uniform	100.0	100.0	91.2	99.6	100.0	10.6	99.9	98.3	98.7	89.7	99.1	87.5	89.7
	MOSAIC	100.0	100.0	97.8	99.6	100.0	92.8	99.8	98.8	99.2	90.8	99.1	98.6	98.0
2048	Uniform	100.0	100.0	96.6	99.4	100.0	48.8	99.9	98.5	99.6	90.9	99.3	92.9	93.9
	MOSAIC	100.0	100.0	99.6	99.8	100.0	97.4	99.8	98.0	99.3	91.9	99.2	99.3	98.6
L2Norm														
128	Uniform	28.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	7.1	15.0	0.6	3.5	4.6
	MOSAIC	90.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	9.0	39.2	11.6	11.3	13.7
256	Uniform	99.6	1.0	0.0	0.0	0.0	0.0	0.0	0.0	18.5	44.9	31.0	12.6	17.7
	MOSAIC	99.0	0.4	0.0	0.0	0.0	0.0	0.0	0.1	16.0	52.9	60.8	12.4	20.8
512	Uniform	100.0	5.6	0.0	0.4	0.2	0.0	0.1	0.1	35.9	64.2	92.4	13.3	27.2
	MOSAIC	100.0	20.2	0.0	1.4	0.4	0.0	0.0	0.0	44.8	63.2	93.6	15.2	29.4
1024	Uniform	100.0	32.0	0.2	5.4	1.4	0.4	0.2	0.2	73.6	79.1	99.0	17.5	35.6
	MOSAIC	100.0	91.2	40.2	82.0	40.4	2.2	64.2	70.8	70.8	78.1	97.4	61.4	67.0
1536	Uniform	100.0	64.0	0.0	16.4	2.0	1.4	2.1	2.8	92.8	84.4	99.2	23.6	42.3
	MOSAIC	100.0	99.6	97.2	97.4	92.4	42.4	88.6	95.2	95.7	85.1	98.4	89.1	90.2
2048	Uniform	100.0	87.0	1.4	38.4	3.8	5.8	12.1	13.2	97.9	87.8	99.2	32.7	49.7
	MOSAIC	100.0	99.8	100.0	99.0	98.2	95.2	96.6	96.5	97.3	87.3	99.3	98.2	97.2

Table 16: **RULER** subtask accuracy (string match, %), uniform vs. MOSAIC per cell, Llama-3-8B-Instruct. AdaKV B64 has only the rescaled winner, compared with uniform-64 from Stage 2.

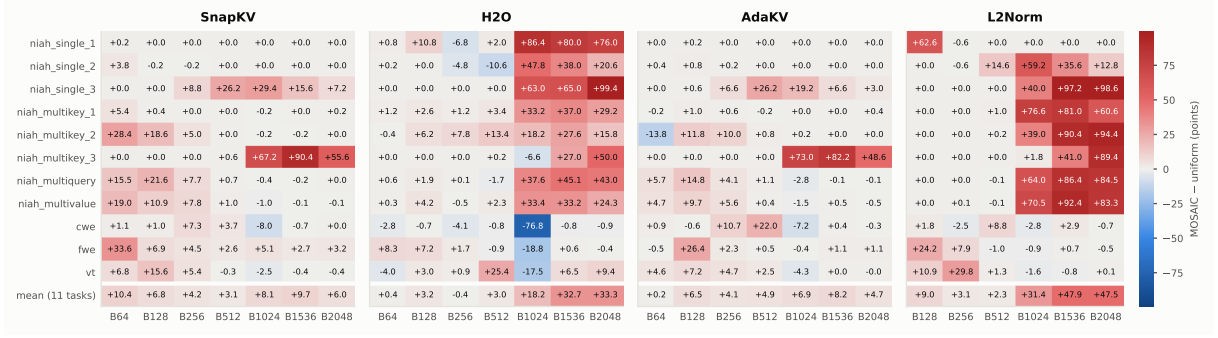


Figure 13: **RULER** subtask gains of MOSAIC over uniform, Llama-3-8B-Instruct (the Δ rows of Table 16, points). Rows: subtasks and their mean; columns: budget. Red: MOSAIC above uniform; blue: below.

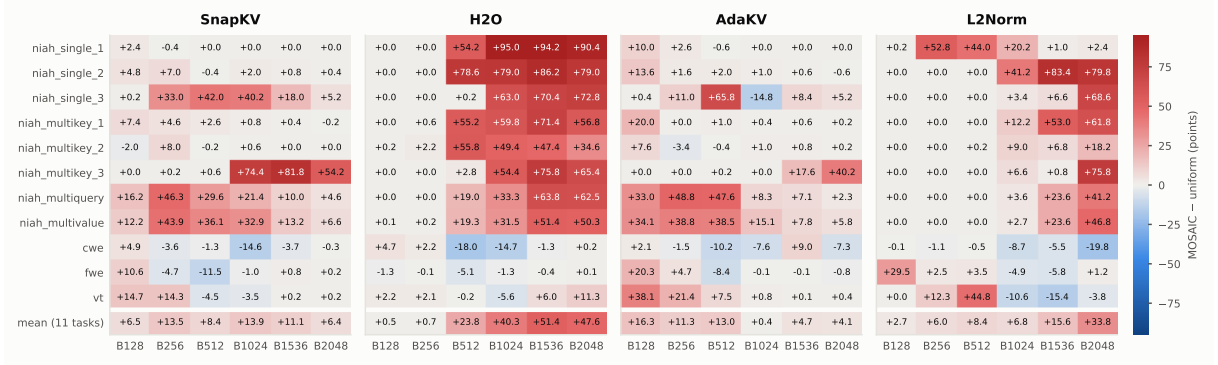


Figure 14: **RULER** subtask gains of MOSAIC over uniform, Mistral-7B-Instruct-v0.2 (the Δ rows of Table 17, points), drawn as Figure 13.

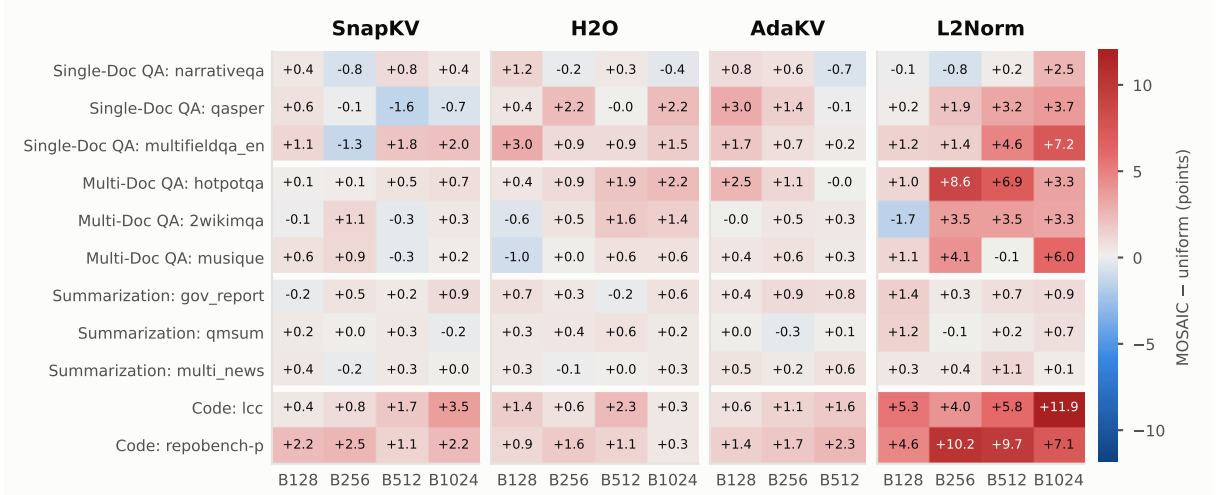


Figure 15: **LongBench** dataset gains of MOSAIC over uniform, Llama-3-8B-Instruct (full data, points; the uniform and MOSAIC rows of Table 18). Rows: datasets grouped by category; columns: budget; grey: no cell. Red: MOSAIC above uniform; blue: below.

Budget		S1	S2	S3	MK1	MK2	MK3	MQ	MV	CWE	FWE	VT	NIAH	All
SnapKV														
128	Uniform	96.8	83.0	0.0	80.4	69.0	0.0	4.7	5.2	8.8	61.3	40.6	42.4	40.9
	MOSAIC	99.2	87.8	0.2	87.8	67.0	0.0	20.9	17.4	13.8	71.9	55.3	47.5	47.4
256	Uniform	100.0	90.4	1.4	90.6	86.6	0.0	14.8	12.9	24.1	78.9	69.3	49.6	51.7
	MOSAIC	99.6	97.4	34.4	95.2	94.6	0.2	61.1	56.9	20.5	74.1	83.6	67.4	65.2
512	Uniform	100.0	95.4	16.8	94.2	94.4	0.0	38.9	25.6	50.3	85.8	89.7	58.2	62.8
	MOSAIC	100.0	95.0	58.8	96.8	94.2	0.6	68.5	61.8	48.9	74.3	85.2	72.0	71.3
1024	Uniform	100.0	97.8	57.0	97.0	99.2	0.0	74.9	58.4	77.6	89.5	97.0	73.0	77.1
	MOSAIC	100.0	99.8	97.2	97.8	99.8	74.4	96.3	91.2	63.0	88.5	93.6	94.6	91.1
1536	Uniform	100.0	99.0	82.0	97.2	99.6	8.8	88.5	80.8	86.7	90.3	98.2	82.0	84.7
	MOSAIC	100.0	99.8	100.0	97.6	99.6	90.6	98.5	94.1	83.0	91.1	98.5	97.5	95.7
2048	Uniform	100.0	99.4	92.4	97.6	99.8	37.4	94.3	90.5	91.1	91.1	98.9	88.9	90.2
	MOSAIC	100.0	99.8	97.6	97.4	99.8	91.6	98.9	97.1	90.7	91.3	99.0	97.8	96.7
H2O														
128	Uniform	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	26.9	76.5	7.2	0.1	10.1
	MOSAIC	0.0	0.0	0.0	0.0	1.2	0.0	0.0	0.1	31.6	75.1	9.4	0.2	10.7
256	Uniform	0.0	0.0	0.0	0.2	3.4	0.0	0.1	0.1	55.0	83.9	12.3	0.5	14.1
	MOSAIC	0.0	0.0	0.0	0.8	5.6	0.0	0.1	0.3	57.2	83.9	14.4	0.8	14.8
512	Uniform	0.4	0.4	0.0	3.0	11.0	0.2	0.6	0.8	75.2	90.5	29.2	2.1	19.2
	MOSAIC	54.6	79.0	0.2	58.2	66.8	3.0	19.6	20.2	57.1	85.4	28.9	37.7	43.0
1024	Uniform	2.8	2.0	0.0	9.8	27.8	3.4	4.5	6.7	88.4	92.1	67.0	7.1	27.7
	MOSAIC	97.8	81.0	63.0	69.6	77.2	57.8	37.8	38.1	73.7	90.9	61.4	65.3	68.0
1536	Uniform	3.8	7.8	0.0	17.4	41.6	12.6	10.9	14.6	92.1	92.7	80.0	13.6	34.0
	MOSAIC	98.0	94.0	70.4	88.8	89.0	88.4	74.8	66.0	90.8	92.3	86.0	83.7	85.3
2048	Uniform	9.6	19.4	0.0	28.0	54.0	27.4	18.8	23.4	93.7	92.9	84.4	22.6	41.1
	MOSAIC	100.0	98.4	72.8	84.8	88.6	92.8	81.2	73.7	94.0	93.0	95.7	86.5	88.6
AdaKV														
128	Uniform	79.8	77.2	0.0	70.8	77.2	0.0	2.4	4.2	11.4	45.4	31.3	38.9	36.3
	MOSAIC	89.8	90.8	0.4	90.8	84.8	0.0	35.4	38.2	13.5	65.7	69.4	53.8	52.6
256	Uniform	97.2	90.4	0.4	91.0	91.0	0.0	8.5	12.2	19.2	76.5	52.8	48.8	49.0
	MOSAIC	99.8	92.0	11.4	91.0	87.6	0.0	57.2	51.0	17.8	81.1	74.2	61.3	60.3
512	Uniform	100.0	95.2	14.2	95.2	95.6	0.0	35.5	32.2	35.3	85.9	81.1	58.5	60.9
	MOSAIC	99.4	97.2	80.0	96.2	95.2	0.2	83.1	70.8	25.1	77.5	88.6	77.8	73.9
1024	Uniform	100.0	98.0	63.4	97.0	97.6	0.0	76.0	65.8	62.0	88.2	95.7	74.7	76.7
	MOSAIC	100.0	99.0	48.6	97.4	98.6	0.0	84.2	80.9	54.5	88.1	96.6	76.1	77.1
1536	Uniform	100.0	99.0	84.2	97.2	98.8	1.4	89.0	83.0	79.6	89.7	97.6	81.6	83.6
	MOSAIC	100.0	99.6	92.6	97.8	99.6	19.0	96.2	90.8	88.6	89.7	97.7	86.9	88.3
2048	Uniform	100.0	99.6	92.8	97.6	99.4	13.0	94.1	90.7	86.9	90.7	98.5	85.9	87.6
	MOSAIC	100.0	99.0	98.0	97.8	99.6	53.2	96.4	96.5	79.6	89.9	98.9	92.6	91.7
L2Norm														
128	Uniform	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	6.7	0.0	0.0	0.6
	MOSAIC	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	36.1	0.0	0.0	3.3
256	Uniform	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.5	44.9	0.0	0.0	4.2
	MOSAIC	53.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.5	47.4	12.4	6.6	10.3
512	Uniform	41.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	7.2	65.8	7.6	5.1	11.1
	MOSAIC	85.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	6.7	69.3	52.4	10.7	19.4
1024	Uniform	79.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	18.1	77.5	76.0	10.0	22.9
	MOSAIC	100.0	41.2	3.4	12.2	9.0	6.6	3.6	2.7	9.5	72.5	65.5	22.3	29.7
1536	Uniform	94.4	0.6	0.0	0.2	0.8	0.0	0.0	0.0	32.7	83.1	91.7	12.0	27.6
	MOSAIC	95.4	84.0	6.6	53.2	7.6	0.8	23.6	23.6	27.1	77.3	76.3	36.9	43.2
2048	Uniform	97.6	0.8	0.0	0.0	2.0	1.0	0.0	0.0	54.8	85.9	96.0	12.7	30.7
	MOSAIC	100.0	80.6	68.6	61.8	20.2	76.8	41.2	46.8	35.0	87.1	92.2	62.0	64.6

Table 17: **RULER** subtask accuracy (string match, %), uniform vs. MOSAIC per cell, Mistral-7B-Instruct-v0.2.

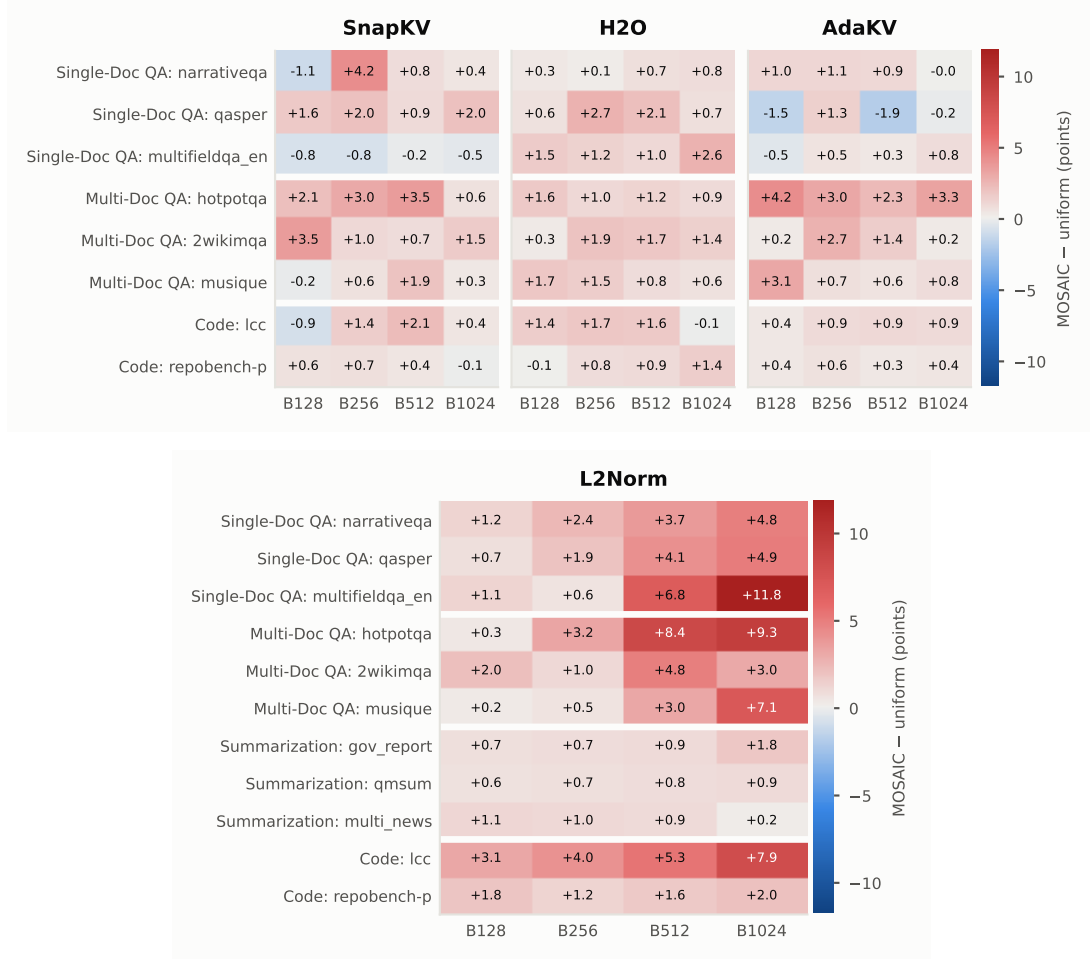


Figure 16: **LongBench dataset gains of MOSAIC over uniform, Mistral-7B-Instruct-v0.2** (full data, points; the uniform and MOSAIC rows of Table 19), drawn as Figure 15; both panels share one colour scale. Top: SnapKV, H2O and AdaKV, which have no Summarization cells; bottom: L2Norm, all four tasks.

Budget		Single-Doc QA				Multi-Doc QA				Summarization				Code		
		NQA	Qasper	MFQA	Mean	HQA	2Wiki	MuSiQue	Mean	GovRep	QMSum	MNews	Mean	LCC	RepoB-P	Mean
SnapKV																
128	Uniform	18.70	35.14	45.08	32.97	45.94	36.39	22.13	34.82	21.01	20.35	21.73	21.03	56.90	52.73	54.81
	MOSAIC	19.07	35.76	46.18	33.67	46.02	36.28	22.77	35.02	20.82	20.54	22.09	21.15	57.25	54.94	56.09
256	Uniform	19.96	39.44	46.35	35.25	46.94	37.50	22.36	35.60	22.25	21.09	23.57	22.30	58.23	53.12	55.67
	MOSAIC	19.21	39.32	45.01	34.51	47.05	38.58	23.30	36.31	22.77	21.09	23.37	22.41	59.05	55.61	57.33
512	Uniform	19.99	41.86	46.06	35.97	46.72	37.98	22.60	35.77	24.06	21.37	24.88	23.44	59.16	54.31	56.73
	MOSAIC	20.83	40.29	47.86	36.33	47.20	37.69	22.29	35.73	24.26	21.69	25.17	23.71	60.89	55.44	58.16
1024	Uniform	20.66	42.64	46.02	36.44	47.18	38.40	22.02	35.87	25.56	22.31	26.67	24.85	58.35	53.69	56.02
	MOSAIC	21.07	41.90	48.05	37.01	47.86	38.73	22.19	36.26	26.49	22.12	26.70	25.10	61.89	55.93	58.91
H2O																
128	Uniform	17.70	31.10	35.84	28.21	43.42	33.38	20.91	32.57	23.02	17.79	24.82	21.88	49.92	42.00	45.96
	MOSAIC	18.86	31.49	38.82	29.72	43.80	32.78	19.91	32.16	23.72	18.10	25.12	22.31	51.33	42.94	47.13
256	Uniform	18.66	32.31	38.91	29.96	44.03	33.42	19.90	32.45	24.78	18.94	25.68	23.13	53.45	45.54	49.50
	MOSAIC	18.44	34.49	39.84	30.92	44.88	33.94	19.94	32.92	25.12	19.32	25.56	23.63	54.04	47.14	50.59
512	Uniform	19.20	35.71	42.32	32.41	43.90	34.29	20.75	32.98	26.95	19.43	26.28	24.22	55.97	49.65	52.81
	MOSAIC	19.49	35.70	43.21	32.80	45.75	35.92	21.32	34.33	26.72	20.03	26.30	24.35	58.23	50.78	54.50
1024	Uniform	19.26	38.36	43.19	33.60	44.33	35.56	20.72	33.54	27.94	20.05	26.93	24.97	59.62	54.23	56.92
	MOSAIC	18.82	40.58	44.68	34.69	46.54	36.94	21.30	34.93	28.52	20.24	27.19	25.32	59.92	54.54	57.23
AdaKV																
128	Uniform	18.60	34.52	45.36	32.83	45.50	36.81	22.67	34.99	21.57	21.26	22.32	21.72	59.83	54.77	57.30
	MOSAIC	19.39	37.49	47.05	34.64	47.98	36.80	23.10	35.96	22.02	21.28	22.86	22.05	60.44	56.13	58.28
256	Uniform	20.09	39.10	46.04	35.08	46.07	37.57	22.63	35.42	22.57	21.81	23.77	22.72	60.02	56.36	58.19
	MOSAIC	20.69	40.46	46.78	35.98	47.18	38.03	23.21	36.14	23.48	21.53	23.99	23.00	61.17	58.08	59.62
512	Uniform	21.32	41.86	46.87	36.68	47.20	38.68	22.53	36.14	23.88	21.86	25.12	23.62	61.09	56.31	58.70
	MOSAIC	20.65	41.74	47.07	36.49	47.19	39.00	22.79	36.33	24.63	21.94	25.68	24.08	62.71	58.65	60.68
L2Norm																
128	Uniform	5.15	15.36	10.10	10.20	11.70	10.32	5.24	9.09	17.95	15.50	19.73	17.73	23.23	21.85	22.54
	MOSAIC	5.02	15.53	11.34	10.63	12.73	8.66	6.31	9.23	19.32	16.72	20.02	18.69	28.55	26.41	27.48
256	Uniform	9.01	18.58	14.15	13.91	18.39	13.17	8.13	13.23	19.85	17.55	21.97	19.79	27.00	20.15	23.57
	MOSAIC	8.24	20.50	15.54	14.76	27.00	16.69	12.22	18.64	20.11	17.49	22.33	19.98	30.99	30.39	30.69
512	Uniform	12.47	22.02	20.95	18.48	31.48	19.41	15.30	22.06	21.40	18.77	23.73	21.30	31.33	24.96	28.14
	MOSAIC	12.65	25.18	25.57	21.13	38.34	22.91	15.22	25.49	22.09	18.96	24.79	21.95	37.15	34.65	35.90
1024	Uniform	16.19	29.28	28.78	24.75	40.64	29.42	16.09	28.72	22.91	19.98	26.14	23.01	37.05	32.85	34.95
	MOSAIC	18.67	32.94	36.01	29.21	43.95	32.68	22.08	32.90	23.80	20.64	26.28	23.57	48.99	39.95	44.47

Table 18: **LongBench dataset scores, uniform vs. MOSAIC per cell, Llama-3-8B-Instruct** (full data); bold: the better of the two in each column. NQA: NarrativeQA; MFQA: MultiFieldQA-en; HQA: HotpotQA; 2Wiki: 2WikiMQA; GovRep: GovReport; MNews: Multi-News; RepoB-P: RepoBench-P.

		Single-Doc QA				Multi-Doc QA				Code		
Budget		NQA	Qasper	MFQA	Mean	HQA	2Wiki	MuSiQue	Mean	LCC	RepoB-P	Mean
SnapKV												
128	Uniform	21.54	21.91	45.38	29.61	33.94	22.04	15.61	23.86	52.20	47.31	49.76
	MOSAIC	20.39	23.49	44.61	29.50	36.03	25.58	15.42	25.68	51.33	47.90	49.61
256	Uniform	22.03	24.99	48.20	31.74	36.62	25.11	15.14	25.62	53.88	50.30	52.09
	MOSAIC	26.26	26.94	47.39	33.53	39.63	26.07	15.73	27.14	55.32	50.97	53.14
512	Uniform	24.13	28.84	48.75	33.91	38.82	25.48	15.10	26.47	55.10	52.44	53.77
	MOSAIC	24.92	29.69	48.55	34.39	42.35	26.22	17.02	28.53	57.23	52.88	55.05
1024	Uniform	25.03	30.37	49.57	34.99	41.84	26.53	18.03	28.80	56.87	53.60	55.23
	MOSAIC	25.40	32.34	49.11	35.62	42.45	28.01	18.33	29.60	57.30	53.46	55.38
H2O												
128	Uniform	22.16	20.57	35.88	26.20	30.65	19.70	12.96	21.10	42.71	36.47	39.59
	MOSAIC	22.48	21.20	37.38	27.02	32.23	19.96	14.67	22.29	44.16	36.40	40.28
256	Uniform	22.23	22.48	39.06	27.92	34.47	20.86	15.78	23.70	46.88	39.30	43.09
	MOSAIC	22.36	25.14	40.26	29.25	35.48	22.77	17.24	25.16	48.59	40.06	44.33
512	Uniform	23.47	26.40	41.70	30.52	35.29	22.41	17.84	25.18	50.81	42.05	46.43
	MOSAIC	24.17	28.51	42.71	31.80	36.47	24.15	18.67	26.43	52.40	42.96	47.68
1024	Uniform	23.44	29.67	43.32	32.14	36.21	25.76	17.76	26.58	54.07	45.23	49.65
	MOSAIC	24.23	30.35	45.93	33.50	37.07	27.12	18.38	27.52	53.97	46.64	50.30
AdaKV												
128	Uniform	21.59	24.37	46.76	30.91	35.63	25.63	14.56	25.27	53.19	48.96	51.08
	MOSAIC	22.63	22.84	46.25	30.57	39.82	25.86	17.66	27.78	53.54	49.31	51.42
256	Uniform	23.87	27.15	48.40	33.14	38.35	25.21	16.41	26.66	54.67	50.64	52.66
	MOSAIC	24.96	28.49	48.92	34.12	41.40	27.87	17.14	28.80	55.57	51.27	53.42
512	Uniform	24.17	30.10	49.00	34.42	39.95	25.37	17.55	27.62	56.02	53.28	54.65
	MOSAIC	25.10	28.20	49.34	34.21	42.23	26.82	18.10	29.05	56.95	53.58	55.27
1024	Uniform	24.86	31.37	48.67	34.97	41.18	27.36	18.03	28.86	56.77	53.38	55.08
	MOSAIC	24.85	31.13	49.48	35.15	44.48	27.61	18.79	30.29	57.68	53.83	55.75

		Single-Doc QA				Multi-Doc QA				Summarization				Code		
Budget		NQA	Qasper	MFQA	Mean	HQA	2Wiki	MuSiQue	Mean	GovRep	QMSum	MNews	Mean	LCC	RepoB-P	Mean
L2Norm																
128	Uniform	3.52	4.35	7.21	5.03	7.33	5.48	2.08	4.96	20.11	15.11	18.84	18.02	24.95	19.94	22.45
	MOSAIC	4.71	5.06	8.35	6.04	7.67	7.53	2.32	5.84	20.83	15.71	19.89	18.81	28.01	21.77	24.89
256	Uniform	6.65	5.71	9.44	7.27	7.64	6.76	2.79	5.73	21.55	16.37	20.90	19.61	28.00	23.16	25.58
	MOSAIC	9.04	7.60	9.99	8.88	10.81	7.77	3.29	7.29	22.29	17.06	21.86	20.40	32.03	24.33	28.18
512	Uniform	9.08	8.36	12.06	9.83	10.96	7.11	4.57	7.55	22.49	17.56	22.62	20.89	31.11	26.25	28.68
	MOSAIC	12.74	12.43	18.82	14.66	19.35	11.91	7.62	12.96	23.34	18.33	23.48	21.72	36.39	27.86	32.12
1024	Uniform	13.29	14.83	20.70	16.27	16.41	10.62	6.00	11.01	23.48	19.17	24.79	22.48	36.59	29.84	33.22
	MOSAIC	18.11	19.76	32.51	23.46	25.67	13.58	13.10	17.45	25.23	20.04	24.95	23.41	44.53	31.83	38.18

Table 19: **LongBench dataset scores, uniform vs. MOSAIC per cell, Mistral-7B-Instruct-v0.2** (full data); bold: the better of the two in each column. NQA: NarrativeQA; MFQA: MultiFieldQA-en; HQA: HotpotQA; 2Wiki: 2WikiMQA; GovRep: GovReport; MNews: Multi-News; RepoB-P: RepoBench-P. SnapKV, H2O and AdaKV (top) have no Summarization cells; L2Norm (bottom) covers all four tasks.